

Search For Spatial Coincidences Between IceCube Neutrinos And Galactic Sources

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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A handwritten signature in blue ink, appearing to read 'Fathima Shifa', with a horizontal line drawn underneath it.

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Approval Sheet

This thesis entitled **Search For Spatial Coincidences Between IceCube Neutrinos And Galactic Sources** by Fathima Shifa. M is approved for the degree of Master of Science from IIT Hyderabad.



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Dedication

I dedicate this to my most supportive and wonderful parents and siblings, and also
to my guide, Prof.Shantanu Desai

Abstract

Multimessenger astronomy is opening up a new field of exploration of the energetic universe through different messengers. One among these messengers is neutrinos, whose origin is still unknown. Many studies have been conducted to search for the potential sources of neutrinos using different detectors.

In this thesis, we searched for the spatial coincidence between high-energy neutrinos detected by the IceCube Neutrino Detector using the publicly available 10-year muon track data with different Galactic sources, which are gamma-ray bright red dwarfs, magnetars, and fast radio bursts. We did not find any spatial coincidence with any of the sources we analyzed, and in the case of magnetars, we also performed a stacked analysis, which similarly did not show any statistically significant excess.

Our work with red dwarfs resulted in a publication in the *Journal of High Energy Astrophysics*, and the analysis of magnetars is currently undergoing the peer-review process.

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Chapter 1

Neutrino Astronomy within the frame work of Multi Messenger Astronomy

For thousands of years, human understanding of the cosmos was limited to observations made with the naked eye. Early astronomers used the stars to track the seasonal changes and began to identify patterns in the motion of the Moon, Sun, and planets.. Then the invention of the telescope during the Renaissance marked a major turning point, opening up new avenues for astronomical discovery.

The fundamental interactions observed in nature include four forces: the electromagnetic force, gravitational force, weak nuclear force, and strong nuclear force, only the electromagnetic force, emitted as optical wavelength photons served as the sole messenger that allowed astronomers to study the distant universe until the middle of the 20th CE. Technological advances later expanded this to include other parts of the electromagnetic spectrum encompasses a wide range of photon energies, extending from low-energy radio waves to high-energy gamma (γ) rays. [ZBPP17]. In recent decades, collaboration between astronomers and physicists has enabled the collection of information from several different sources or messengers, not limited to electromagnetic signals but also including neutrinos and gravitational waves. This advancement has enabled the comprehensive utilization of all four fundamental interactions in nature, thereby uncovering novel and intriguing features of the universe and its underlying components. [MFHM19].

Multi-messenger astronomy offers a revolutionary perspective on exploring and un-

derstanding the universe, in which information about astronomical events is collected using various types of messengers [Sha24]. Combining the information coming from messengers promises to help solve the still unanswered questions on cosmic-ray sources and acceleration mechanisms.

1.1 Multi Messengers

The universe is continuously traversed by elementary and subatomic particles that travel through space at extremely high velocities. Many of these particles reach Earth, carrying invaluable information about the astrophysical phenomena that generated them. These so-called *cosmic messengers* are currently the only means by which we can probe the distant and energetic processes of the Universe, of which we are a part.

A significant milestone in contemporary astrophysics was reached on September 14, 2015, when the Laser Interferometer Gravitational-Wave Observatory (LIGO) made the first direct observation of gravitational waves. [AAA+16]. This signal originated from the merger of two black holes in a binary system. The detection not only initiated the era of gravitational-wave astronomy but also highlighted the increasing significance of multi-messenger techniques in probing the cosmos. Although gravitational waves are still far from becoming a dominant observational tool, placing them among the key cosmic messengers, in addition to high-energy photons, neutrinos, and charged cosmic rays, offers a powerful means to investigate the highly energetic and mysterious processes occurring in the cosmos. The multi-messenger approach is depicted in 1.1. The combination of information from these diverse messengers holds great promise for addressing longstanding questions about the astrophysical sources and underlying mechanisms that drive the acceleration of cosmic rays.

Among these cosmic messengers, neutrinos hold a crucial position due to their extremely weak interactions with ordinary matter, allowing them to escape dense astrophysical environments and travel unimpeded across the Universe. The IceCube Neutrino Observatory contributes to this global multi-messenger effort through its Real-Time Alert System. The main aim of such follow-up observations is to pinpoint the astrophysical origins of the observed neutrino events.

IceCube plays a significant role in the Astrophysical Multimessenger Observatory Network (AMON), a collaborative initiative coordinated by Pennsylvania State Uni-

versity [SFC⁺13]. AMON aims to enable real-time correlation analyses across various high-energy signals, including photons, neutrinos, cosmic rays, and gravitational waves. The AMON network also includes several prominent observatories such as ANTARES, the Pierre Auger Observatory—a major ground-based facility for detecting cosmic rays—*Fermi*, a space-based gamma-ray telescope, the High Altitude Water Cherenkov (HAWC) Observatory, which observes gamma rays and cosmic rays in the 100GeV to 100TeV energy range, and the Laser Interferometer Gravitational-Wave Observatory (LIGO), known for detecting gravitational waves.

Before narrowing our focus to neutrinos and the potential astrophysical sources from which we expect them to originate, we shall first provide a brief description of the critical messengers in modern astronomy: cosmic rays.

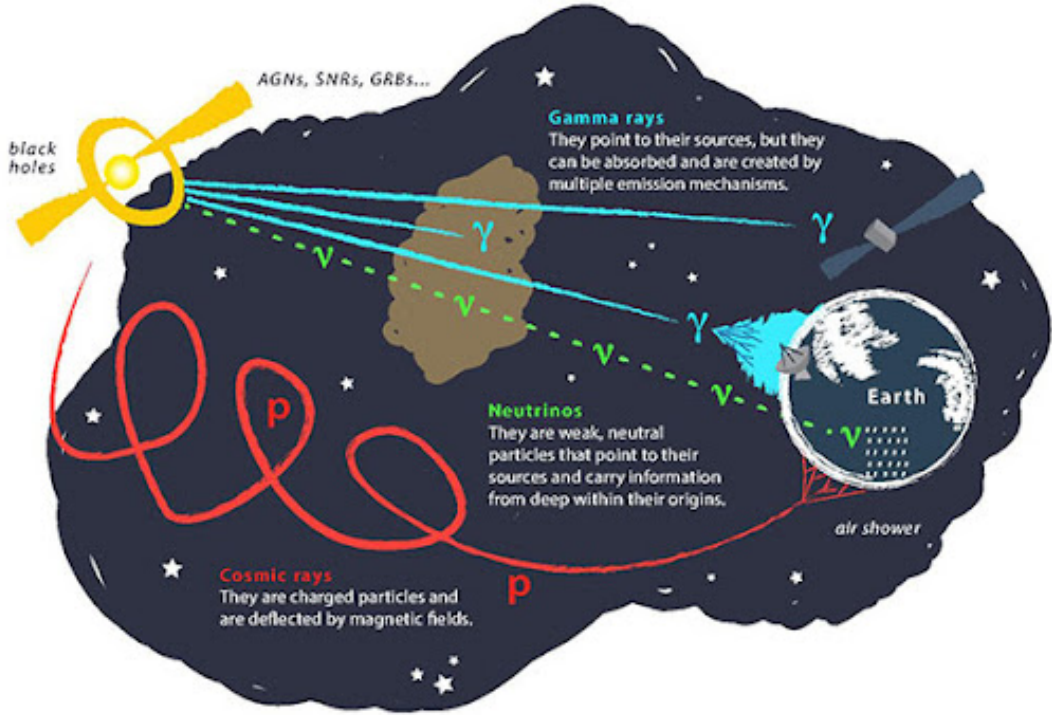


Figure 1.1: Illustration of multi-messenger astronomy. Astrophysical sources can produce cosmic rays (CRs), photons, and neutrinos (note: gravitational waves are not shown). As they travel, cosmic rays are deflected by magnetic fields before reaching Earth, where they are observed using air shower detectors. High-energy photons can be detected by both space-based and ground-based telescopes, although their propagation may be hindered by interactions with dust or low-energy background radiation. In contrast, neutrinos can travel directly from their source to Earth without being absorbed. Image credit: <http://www.icehap.chiba-u.jp/en/multimessenger/index.html>

1.1.1 Cosmic Rays

Victor Hess first identified cosmic rays in 1912 through measurements taken during a high-altitude balloon flight. where he observed an increase in atmospheric ionization with altitude, indicating the presence of high-energy radiation from outer space. These rays are high-energy ionized nuclei—mostly protons (90%), alpha particles (9%), and heavier nuclei—that travel at relativistic speeds, with a small fraction reaching ultra-relativistic energies up to 10^{20} eV

To investigate their origin, scientists analyze their composition, energy spectrum, and arrival directions, comparing them with known astrophysical sources, including the Sun, supernova remnants, and neutron stars. The cosmic-ray energy spectrum exhibits a power-law behavior. [1.2](#), showing key features like the 'knee' around $\sim 10^{15}$ eV, where the slope of the spectrum becomes steeper due to limitations in galactic acceleration or confinement, and the “ankle” at $\sim 10^{18}$ eV, indicating a transition to extragalactic sources [\[GER16\]](#). Detection methods include direct measurements via balloon and satellite experiments for lower energies, and indirect observations of extensive air showers for higher energies. These studies reveal that cosmic rays are likely accelerated by strong magnetic fields, particularly in supernova remnants, with the maximum energy they can attain is limited by the particles' gyroradius and the magnetic field strength at the source.

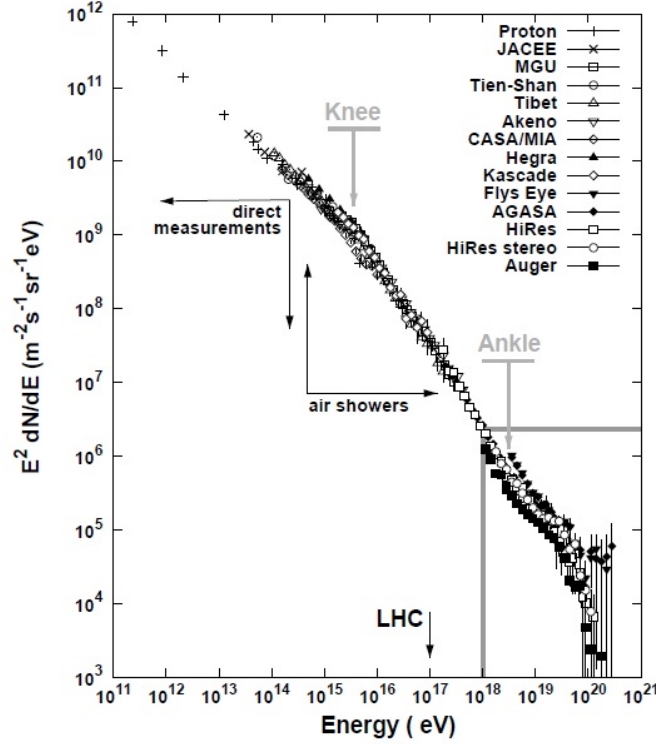


Figure 1.2: The plot presents the differential energy spectrum of cosmic rays above 10^{11} eV, scaled by E^2 to highlight spectral features. Gray arrows mark the locations of the 'knee' and 'ankle' in the spectrum. The figure also indicates the energy ranges accessible through direct detection techniques and extensive air shower observations, along with the corresponding laboratory energy scale of the Large Hadron Collider.

The elemental makeup of cosmic rays, when contrasted with that of the solar system [1.3](#), reveals both parallels and significant differences. Both compositions display the odd–even effect, where nuclei possessing even atomic numbers (Z) tend to be more prevalent, attributed to their relatively greater nuclear binding energies. Nonetheless, two main distinctions are evident. First, nuclei with atomic number $Z > 1$ appear in cosmic rays at a notably higher ratio compared to protons than what is observed in solar system matter. The underlying reason for this disparity is not yet fully understood; it could stem from the challenge of ionizing hydrogen efficiently for acceleration processes or may indicate intrinsic differences in the sources themselves. Second, elements such as lithium, beryllium, and boron (Li, Be, B), along with scandium, titanium, vanadium, chromium, and manganese (Sc, Ti, V, Cr, Mn), are detected in cosmic rays at abundances vastly exceeding their solar system levels. These nuclei are predominantly produced by spallation reactions, wherein heavier elements like carbon, oxygen, and iron undergo fragmentation upon collisions with the interstellar medium (ISM). The elevated presence of these secondary nuclei serves as

a critical diagnostic tool for understanding the transport and retention properties of cosmic rays throughout the Galaxy.

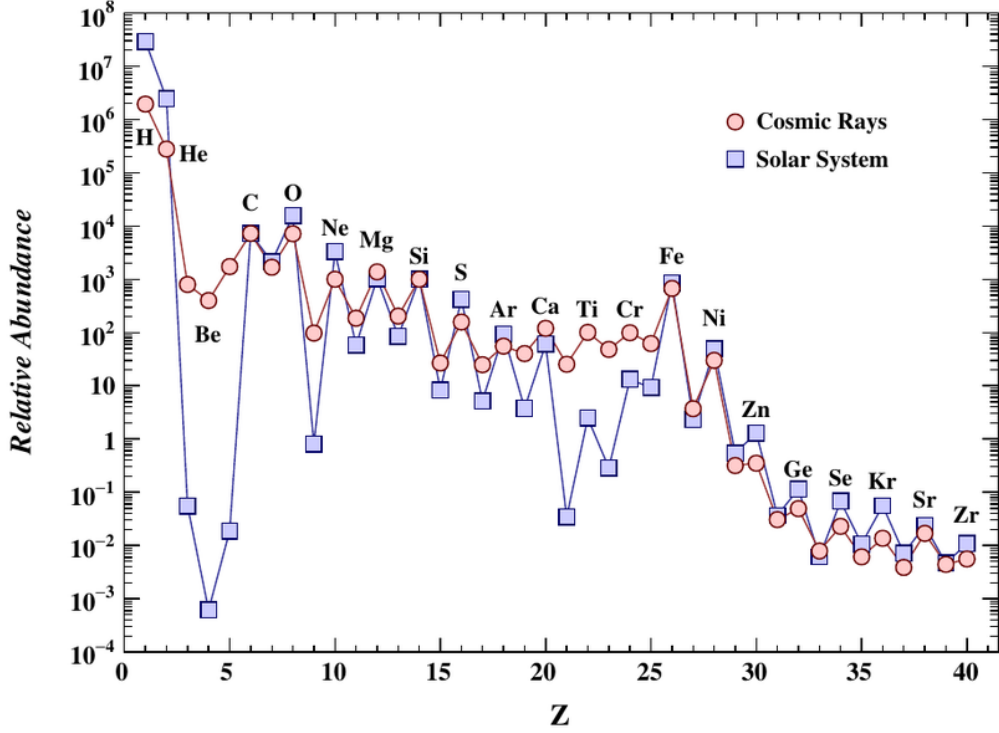


Figure 1.3: Elemental abundances of cosmic rays detected on Earth (shown as filled symbols with solid lines) compared to solar system abundances (represented by open symbols), all normalized to carbon = 100.

1.2 Multi-Messenger Observatories and Detectors

The investigation of compact objects emitting high-energy radiation began to gain considerable momentum in the 1950s, after a gradual progression during earlier decades facilitated by advancements in ground-based optical telescopes. Significant progress was made with the introduction of large-scale radio telescopes, followed by the launch of space-based observatories equipped with X-ray and gamma-ray detectors. These technological innovations led to the identification of various energetic astrophysical phenomena, including neutron stars, black holes, and active galactic nuclei. They also enabled the detection of explosive and transient events such as X-ray novae, X-ray bursts, and gamma-ray bursts (GRBs).

By the end of the 1960s, large-scale underground detectors were developed to ob-

serve neutrinos, initially capturing signals from the Sun and later from supernova events. A significant breakthrough was achieved with the establishment of the IceCube Neutrino Observatory at the South Pole. Over the past decade, IceCube has successfully detected high-energy neutrinos originating outside our galaxy, in the TeV to PeV energy range. These detections have opened new avenues for probing the distant cosmos, allowing scientists to associate neutrino emissions with astrophysical sources such as blazars and gamma-ray bursts.

Measurements of cosmic rays in the GeV energy range began as early as the 1910s. However, it wasn't until the 1960s that large-scale detectors were capable of probing higher-energy cosmic rays, pointing to their possible extragalactic origins. In more recent times, it has become feasible to study both the energy spectrum and particle composition in the ultra-high-energy range of 10¹⁸–10²⁰ eV. The development of gravitational wave detectors commenced in the 1970s, but meaningful progress only occurred in the 1990s with the introduction of advanced technologies and the construction of sufficiently large arrays. These efforts culminated in the first confirmed detections in 2015 [AAA⁺16]. For the first time in history, observational instruments sensitive to all four fundamental forces of nature have been deployed to investigate the universe, uncovering phenomena that were previously hidden from view 1.4

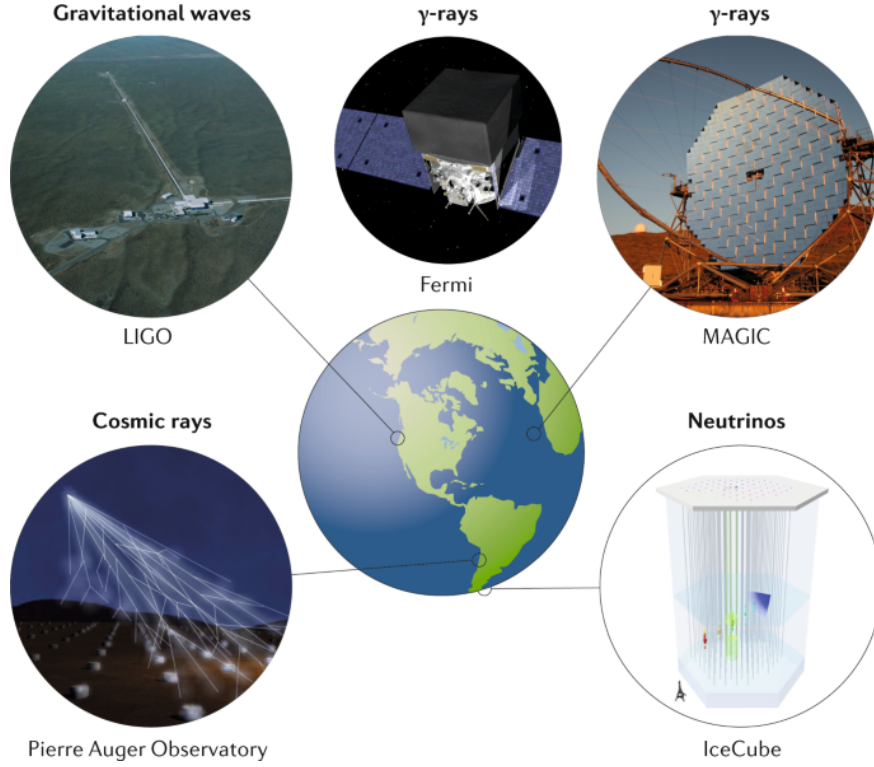


Figure 1.4: Illustration of instruments used to detect cosmic messengers associated with the four fundamental forces: gravitational, electromagnetic, weak, and strong. Clockwise from the top left: the LIGO Hanford interferometer for detecting gravitational waves; the Fermi space telescope for observing gamma rays; one of the MAGIC telescopes designed to detect Cherenkov radiation from high-energy particles; a schematic representation of the IceCube Neutrino Observatory located at the South Pole; and a diagram of the Pierre Auger Observatory in Malargüe, Argentina, used for studying ultra-high-energy cosmic rays.

1.2.1 Astronomy with Optical Telescope

The invention of the telescope by Galileo marked the beginning of astronomical exploration using visible light photons. From this information, the universe known to man for millions of years is the universe with stars and galaxies.

Modern optical astronomy has greatly advanced from Galileo’s early work, now utilizing telescopes with apertures as large as 10 meters—such as the Very Large Telescope (VLT) in Chile—and sophisticated imaging systems that replace the human eye. A key development in this field has been the adoption of Charge-Coupled Devices (CCDs), similar to the sensors found in smartphones, which enable direct digital recording of sky images. These CCDs have facilitated large-scale digital sky mapping

projects, most notably the Sloan Digital Sky Survey (SDSS). The increased aperture size of modern telescopes allows them to collect more light from faint and distant objects, thereby improving sensitivity and the ability to detect low-flux sources.

1.2.2 Radio Astronomy

During the 20th century, astronomers began exploring the universe using radio waves, giving rise to radio astronomy, expanding our view of the Universe beyond visible light. Using radio telescopes, scientists can observe celestial phenomena within the frequency range of 0.1 to 100 GHz. This range has unveiled a variety of objects that are not characterized by blackbody radiation but instead exhibit emission following a power-law spectrum. These discoveries have broadened our understanding of astrophysical processes and introduced entirely new classes of cosmic sources.

$$\frac{dN_\gamma}{dE_\gamma} \propto E^{-\Gamma_\gamma} \quad (1.1)$$

where Γ denoting the slope of the power-law (or “spectral index”). And we found there is a hidden universe, radiating at wavelengths and frequencies we can’t see with our eyes.

A new era in astronomical observation began in 1930, during the early career of Karl G. Jansky, an engineer working at Bell Laboratories unintentionally discovered radio waves coming from space [ZKL21]. The first professional radio telescope was constructed in the late 1930s by amateur radio astronomer Grote Reber, and the discipline of radio astronomy was developed since Jansky’s 1933 finding in Washington, DC.

The most basic and conventional type of radio telescope is the single-dish model. It is made up of a sizable reflector in the form of a dish which collects incoming radio signals and focuses them onto a receiver positioned at its focal point. The sensitivity and resolving power depend on dish size. Then the development of radio interferometers by radio astronomers allowed them to overcome the constraints of single-dish telescopes. These devices are operated as an array and made up of numerous smaller radio antennae dispersed over large distances. They produce substantially higher resolution and sensitivity by combining the signals from each antenna, which is similar to building a virtual telescope with a size equal to the maximum distance between

antennae. Very-Large Array (VLA) in USA 1.5b and Atacama Large Millimeter/sub-millimeter Array (ALMA) in Chile 1.5a are two prominent interferometric arrays. After combining signals from different antennas, we can generate interference patterns and thereby reconstruct detailed images of celestial objects.

Fundamental parts of a radio telescope are:

- **Reflector** – where the incoming radio waves are captured and focused.
- **Receiver** – the concentrated radio waves are transformed into electrical signals that may be analyzed and processed.
- **Data processing system** – to obtain astronomical data from the receivers, data collection, processing, and analysis are required.

As a result, precise radio maps and spectra of the detected celestial objects are produced using sophisticated computer systems and algorithms that clean and interpret the signal.

In modern radio astronomy, advancements are:

- **Aperture synthesis** – in which radio astronomers use interferometry to combine signals from multiple antennas, creating a large virtual aperture that enhances the resolving power of radio telescopes. This allows for detailed observations of distant galaxies and star-forming regions.
- **SDR technology** – which digitizes signals right at the antennas. This adaptability enables more complex data processing, simpler calibration, and effective use of the telescope in a variety of observation modes.
- **Big data and machine learning, receiver technology, array configuration.**

1.2.3 X-ray Astronomy

The study of energetic phenomena—including X-ray emitters, supernova explosions, radio galaxies, and quasars—centers on understanding the processes responsible for producing high-energy photons, especially in the X-ray and gamma-ray portions of the electromagnetic spectrum. This area of research is known as high-energy astrophysics. Unlike radio waves and visible light, X-ray photons cannot penetrate Earth's



(a) VLA ,image source:<https://public.nrao.edu/gallery/vla-panorama/>



(b) ALMA ,image source:<https://www.almaobservatory.org/en/about-alma/>

Figure 1.5: Images of two prominent radio telescopes: the Very Large Array (VLA) located in the United States, and the Atacama Large Millimeter/submillimeter Array (ALMA) situated in Chile.

atmosphere because they are absorbed before reaching the surface.

Unlike visible light and radio waves, X-ray photons are absorbed by Earth’s atmosphere necessitating the deployment of telescopes in space to observe cosmic X-ray sources. In 1962, a pivotal moment occurred when a team led by Riccardo Giacconi launched a rocket equipped with an X-ray detector, resulting in the identification of Scorpius X-1, the very first X-ray source found beyond our solar system.

In the following decades, specialized satellites for X-ray astronomy—such as Uhuru (1970), Ariel 5, SAS-3, OSO-8, and HEAO-1—were launched, greatly enhancing our knowledge of high-energy processes in the cosmos.

Given that X-ray wavelengths are

$$\lambda = E_{\gamma}^{-1} \approx 10^{-7} \left[\frac{1 \text{ keV}}{E_{\gamma}} \right] \text{ cm} \quad (1.2)$$

Given that X-ray wavelengths are comparable to atomic dimensions, traditional refractive optics are ineffective for focusing X-rays. Instead, X-ray telescopes employ grazing incidence optics [1.6a](#), in which X-rays reflect off mirror surfaces at shallow angles, allowing them to be focused onto detectors. Modern X-ray observatories such as Chandra and XMM-Newton [1.6](#) utilize this technology, while future missions such as Athena plan to implement silicon pore optics to achieve larger apertures and improved sensitivity. Gamma rays, which have energies surpassing those of X-rays,

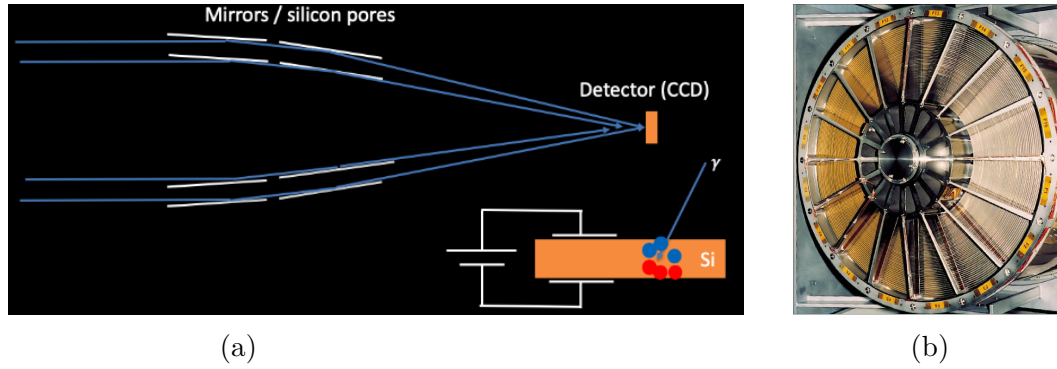


Figure 1.6: The figure(a) shows the grazing incidence optics principle of X-ray telescopes, and the figure(b) shows the X-ray multi-mirror mission-Newton (XMM-Newton), image source: <https://www.cosmos.esa.int/web/xmm-newton/technical-details-mirrors>.

are similarly blocked by Earth's atmosphere, making it necessary to use space-based instruments for their observation.

1.2.4 Gamma-ray Astronomy

Most gamma rays originating from space are absorbed by Earth's atmosphere. This led to the emergence of gamma-ray astronomy [ZNP17] truly advanced when instruments were placed at high altitudes, such as on balloons or satellites, to observe these high-energy photons.

The Explorer 11 satellite, launched in 1961, carried the first gamma-ray telescope to be sent into orbit. It detected fewer than 100 cosmic gamma-ray photons coming from every direction. This suggested the existence of a widespread cosmic gamma-ray background. The earliest significant gamma-ray signal originating from inside our galaxy was detected by an instrument aboard the OSO-3 satellite.

A major breakthrough occurred during the late 1960s and early 1970s, thanks to defense satellites. These *Vela* satellites, initially built to detect gamma-ray flashes from nuclear detonations, unexpectedly picked up intense bursts of gamma radiation referred to as *gamma-ray bursts* (GRBs)—lasting anywhere from less than a second to several minutes and coming from apparently random points in the sky.

The 1991 deployment of the Compton Gamma Ray Observatory (CGRO) marked a significant leap in gamma-ray studies. With four main experiments onboard, CGRO

greatly enhanced the ability to pinpoint and time-resolve gamma-ray sources.

To enable rapid localization of GRBs and allow multi-wavelength follow-up in ultraviolet, visible, and X-rays, NASA deployed the *Swift* satellite in November 2004.

At energies much higher than a few MeV, the cross-section for Compton scattering diminishes and becomes smaller than that for electron-positron pair production. This method is employed in pair-conversion telescopes such as the Fermi Space Telescope [TWH22], where incoming gamma rays are transformed into electron-positron pairs., whose paths are tracked and whose energies are measured using a calorimeter, allowing scientists to determine the incoming gamma ray’s energy and origin—whether from neutron stars, dark matter interactions, or other sources.

This method is employed in pair-conversion telescopes such as the Fermi Space Telescope [TWH22], where incoming gamma rays are transformed into electron-positron pairs.

1.2.5 Very High Energy Gamma-ray Detection Techniques

Gamma rays play a crucial role as observational tools in the study of high-energy astrophysics. Because these photons originate from extraterrestrial sources, their direct detection is only feasible above the Earth’s atmosphere, employing instruments placed on satellites or high-altitude balloons [BCMA21]. The Fermi Gamma-ray Space Telescope detects gamma rays with energies extending up to several hundred GeV through the mechanism of pair production. At such elevated energies, the observed gamma-ray spectrum follows a power-law distribution, indicating that photon counts sharply decline as energy increases. Consequently, detecting these rare high-energy gamma rays necessitates the use of detectors with large collection areas combined with extended observation periods [SHK22].

A primary objective of gamma-ray astronomy is to identify the sources and mechanisms responsible for the production and acceleration of cosmic rays to energies far exceeding those achievable by terrestrial accelerators. Gamma rays with energies above 100 GeV interact with atmospheric nuclei, initiating a cascade of secondary particles referred to as an Extensive Air Shower (EAS). These secondary particles predominantly consist of electrons and positrons, which are typically absorbed within

the atmosphere. However, when the incident gamma-ray energy surpasses approximately 100 TeV, a fraction of these particles can penetrate through the atmosphere and reach the Earth’s surface [20224]. The detection of such particles at ground level constitutes the field of ultra-high-energy (UHE) gamma-ray astronomy.

EAS particles also generate *Cherenkov radiation*, which is emitted in a forward-directed cone and can reach ground-based detectors. For gamma rays with energies spanning from several tens of GeV up to several tens of TeV, even though the particle cascade does not hit the ground, the Cherenkov light produced can be measured. This method is known as the *atmospheric Cherenkov technique*, where Earth’s atmosphere effectively acts as the detector medium for observing very high-energy gamma rays.

1.2.6 Gravitational Wave Astronomy

Gravitational waves, sometimes called gravitational radiation, were initially predicted by Albert Einstein as an outcome of his general theory of relativity. This framework explains gravity as the distortion of spacetime produced by mass and energy. Similar to photons and neutrinos, gravitational waves carry information about their cosmic sources and propagate in straight lines, allowing us to trace them back to their origin. The wavelengths of gravitational waves emitted by astrophysical sources are set by the typical scales of the gravitational dynamics involved. In contrast to electromagnetic waves, gravitational waves are inherently macroscopic.

Due to their extremely weak interaction with matter, gravitational waves pass through material with minimal disturbance. As a result, These waves can only be detected when originating from extremely dense and massive objects, like merging black holes or neutron stars. These systems, which concentrate large amounts of mass in small volumes, can generate strong gravitational radiation. Consequently, the shortest possible wavelength of such gravitational waves is approximately $\lambda_{\text{GW}} \approx 1 \text{ km}$, implying that detectors must be built on very large scales to be effective.

A century following the development of Einstein’s theory of general relativity, the Laser Interferometer Gravitational-Wave Observatory (LIGO) made history in 2015 by directly detecting gravitational waves for the first time. These waves originated from the merger of two black holes [AAA+16], marking the dawn of gravitational

wave astronomy and representing a major milestone in the era of multi-messenger astrophysics.

1.3 Neutrino Astronomy

Traditional astronomy explores the universe spanning the entire electromagnetic spectrum, from radio waves to gamma rays. Observations at these varied wavelengths have greatly enhanced our knowledge of astrophysical phenomena., They fall short at extremely high energies, where photons are absorbed or scattered.

Significant progress in high-energy neutrino astronomy was achieved through the development of the IceCube Neutrino Observatory. This facility employs one cubic kilometer of clear Antarctic ice beneath the South Pole, making it the world’s largest neutrino detector. IceCube has detected neutrinos with energies spanning from tera-electronvolts (TeV) to peta-electronvolts (PeV) originating from extragalactic sources, opening a new chapter in observational astronomy [BDD⁺08a].

1.4 The Ghostly Messengers of the Universe: Understanding Neutrinos

Neutrinos are extremely light, uncharged particles that move at speeds close to that of light. Due to their extremely weak interaction with matter and insensitivity to magnetic fields, they are capable of traversing immense cosmic distances without deflection. This property makes them excellent messengers for studying remote and energetic astrophysical phenomena.

Neutrinos come in three unique varieties, often called “flavors,” each linked to a charged lepton with an electric charge of (-1) : the electron, muon, and tau. These flavors include:

- Electron neutrino
- Muon neutrino
- Tau neutrino

Additionally, every flavor has a corresponding particle known as an antineutrino.

1.5 Neutrino Origins and Discovery

Wolfgang Pauli first proposed neutrinos in 1930 to explain the continuous energy distribution of electrons detected during beta decay, a phenomenon initially observed by Chadwick in 1914. If beta decay involved only two particles, the emitted electrons would have a discrete energy. To reconcile this discrepancy and preserve the law of energy conservation, Pauli postulated the existence of a light, neutral particle with spin- $\frac{1}{2}$, which he initially referred to as the “neutron.” This was later renamed “neutrino” by Enrico Fermi.

Neutrinos interact so weakly with matter that detecting them proved challenging. The electron neutrino was first directly detected in 1956 by Fred Reines and Clyde Cowan. Later, in 1962, Leon Lederman, Melvin Schwartz, and Jack Steinberger identified the muon neutrino. The tau neutrino was observed for the first time in 2000 by the DONUT experiment at Fermilab. Each neutrino flavor is associated with a corresponding charged lepton.

Neutrinos generated through nuclear fusion processes in the sun’s core, known as solar neutrinos, were later detected. However, the observed flux was significantly lower than theoretical predictions. This inconsistency became known as the “solar neutrino problem.” This issue was resolved through the discovery of neutrino oscillations—the phenomenon where neutrinos switch between flavors as they propagate. Experiments like Super-Kamiokande and the Sudbury Neutrino Observatory (SNO) provided the first confirmation that neutrino flavor states do not align with their mass states.

1.5.1 Neutrino Detectors

Neutrino detectors are massive and highly sensitive instruments situated underground, underwater, or within polar ice to reduce background noise from cosmic rays. These detectors typically use materials like water, ice, or liquid scintillators to observe Cherenkov radiation—faint light flashes created when neutrinos interact with atomic nuclei.

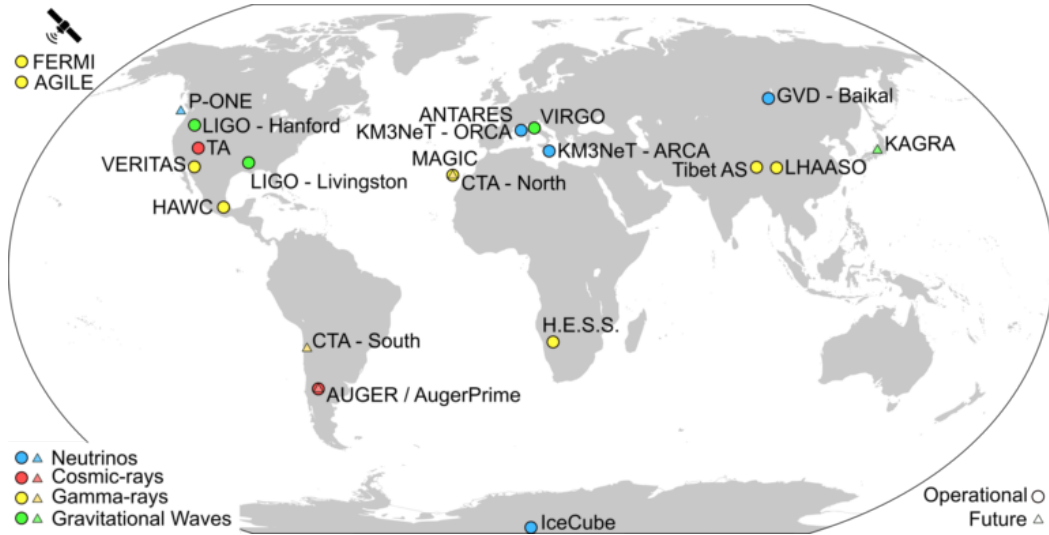


Figure 1.7: This map highlights the locations of major neutrino observatories around the world. These detectors, such as IceCube in Antarctica, Super-Kamiokande in Japan, and KM3NeT in the Mediterranean Sea, are designed to study neutrinos. Image source: <https://km3net.ific.uv.es/km3net/research/neutrino-astrophysics/>

Further details on the methods and instrumentation used in neutrino detection will be discussed in later chapters.

Chapter 2

Probing the Universe with Neutrinos: The IceCube Neutrino Observatory

When the neutrino was observed in 1956 [[Rio01](#)], It was considered an ideal messenger since it could travel without being absorbed or diverted by magnetic fields. Due to the weak interaction of neutrinos, it is difficult to detect them. By 1970, It became evident that detecting cosmic neutrinos generated by cosmic ray interactions with the cosmic microwave background would require a detector on the scale of a cubic kilometer.

It is difficult to build a neutrino telescope of this size. Early efforts focused on converting natural water into a Cherenkov detector, which produces light when a neutrino collides with nuclei inside or close to the detector. The DUMAND project [[You95](#)], built in the sea off Hawaii, the project ended unsuccessfully after twenty years. Nonetheless, it laid the groundwork for various detector technologies, influencing experiments like the Baikal Lake neutrino detector and those located in the Mediterranean Sea.

The vision initially proposed by the DUMAND collaboration was ultimately achieved—not in ocean water, but by utilizing the exceptionally clear, deep Antarctic ice to build a large-volume neutrino detector. This led to the development of AMANDA, which operated between 2000 and 2009. AMANDA served as a successful prototype, paving the way for the construction of the much larger IceCube Neutrino Observatory [[HK10](#)].

2.1 Neutrino Detection: A Historical Trajectory

Efforts to detect neutrinos originating beyond Earth began in the 1960s, with early experiments conducted deep underground at two sites: the Kolar Gold Fields in India and the East Rand mine in South Africa. Here, Neutrino detectors were constructed underground to shield them from the intense background radiation at the Earth's surface, primarily caused by cosmic ray interactions in the atmosphere. By 1967, Ray Davis had developed a geochemical experiment that successfully detected a few argon atoms per day, produced through interactions between solar neutrinos and a large underground tank filled with perchloroethylene.

By 1980, the scintillator detector had advanced to the large MACRO detector [AAA+01] at the Gran Sasso underground laboratory in Italy, approximately 78 meters in length, 2 meters in width, and 9 meters in height. The MACRO experiment incorporated alternating layers of passive absorber material and streamer tubes for tracking, all enclosed by a 12-meter-long tank containing liquid scintillator to detect passing particles through light emission. Over six years, it detected more than 1000 neutrinos.

Around the same time, The Fréjus experiment investigated the spectrum of atmospheric muon neutrinos and established upper bounds on the flux of neutrinos with energies in the TeV range originating from cosmic sources. But to make further progress, a new method was originally proposed by Markov in 1960, These detectors identify charged particles by capturing the Cherenkov radiation they emit as they travel faster than the speed of light in water or ice.

Cherenkov radiation occurs when a charged particle travels through a medium at a velocity exceeding the phase velocity of light in that medium. This effect is similar in concept to a sonic boom, but it involves electromagnetic waves rather than sound. The emitted light, primarily in the blue to near-ultraviolet range, can be captured using photomultiplier tubes (PMTs). By deploying a sufficient number of PMTs within a detection volume, it becomes possible to register neutrinos with energies as low as a few MeV. This detection method was first used in large-scale, kiloton-class detectors designed to observe low-energy neutrino interactions.

The IMB and Kamiokande detectors stand out as two of the most successful early neutrino observatories. Both detectors featured large tanks filled with ultra-pure water,

with photomultiplier tubes (PMTs) installed along the walls and ceilings to capture Cherenkov light., even though they were designed for GeV neutrinos. The achievements of the initial detectors paved the way for more advanced second-generation experiments. Super-Kamiokande, a much larger detector holding 50,000 tons of water, expanded upon Kamiokande’s design [F⁺02], while the Sudbury Neutrino Observatory (SNO) utilized 1,000 tons of heavy water. These two experiments provided crucial evidence that neutrinos possess mass by detecting flavor oscillations in both solar and atmospheric neutrinos, marking the first clear indication of physics beyond the Standard Model.

2.2 From AMANDA to IceCube

The detection of neutrinos using Antarctic ice began in the 1990s with the AMANDA collaboration, which pioneered the use of deep glacial ice as a detection medium. AMANDA was deployed within the 2800m thick Antarctic ice sheet at the South Pole. The deployment involved drilling deep holes using hot water and lowering strings of optical modules into the ice before the water refroze. During the 1993–94 season, the team lowered 80 PMTs into a hole over 1 km deep. The majority of the optical sensors remained operational despite the intense pressure exerted by the re-freezing ice.

However, they initially couldn’t detect cosmic ray muon tracks due to tiny air bubbles about 50 microns in size that were present at 1 km depth. After they confirmed that below 1400m the pressure caused bubbles to collapse, improving the optical clarity of ice, four strings of optical detectors were deployed at depths ranging from 1500 to 2000 deep during the 1995–96 season. They also developed algorithms to reject background noise, and by 2000 the complete AMANDA with 19 strings and 677 modules was successfully detecting around 1000 neutrino events per year [HK10].

Still, AMANDA had many drawbacks: it was too small for large-scale neutrino detection and necessitated labor-intensive recalibration each year. The optical modules contained 8-inch photomultiplier tubes (PMTs) that transmitted analog signals to the surface using cables. These long cable runs degraded signal quality.

Later attempts replaced electrical transmission with an analog optical system using LEDs controlled by PMT output to send signals through optical fibers. This

improved timing but had a narrow dynamic range and suffered from reliability issues over time, requiring frequent recalibration.

These limitations made it clear that AMANDA couldn't be scaled up. The team began investigating alternative approaches, with the most promising yet difficult advancement being the integration of digitized electronics directly within each optical module. To evaluate this, AMANDA's string 18 was equipped with prototype digitizers inside the modules. These digitizers operated alongside the existing fiber-optic analog signal transmission, enabling simultaneous testing of the new technology while maintaining compatibility with the AMANDA system. The results were successful, and this technology was later implemented in the design of the next-generation IceCube detector.

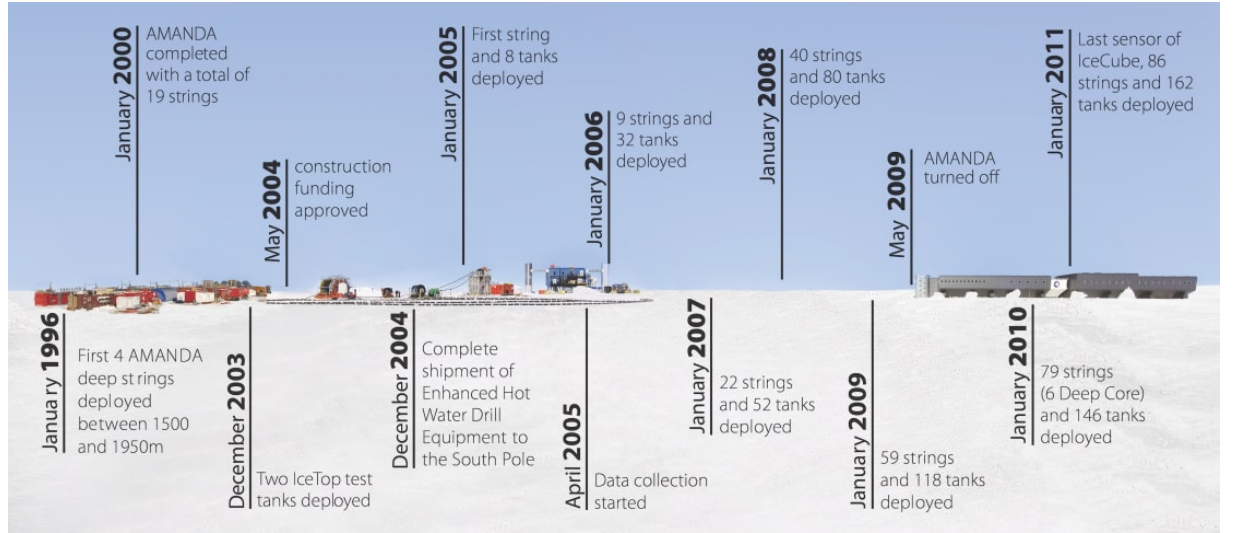


Figure 2.1: Image source: <https://icecube.wisc.edu/science/icecube/>

2.3 Design of the IceCube Detector

The IceCube Neutrino Observatory comprises a network of Digital Optical Modules (DOMs) deployed deep into the Antarctic ice. Among its components are the densely instrumented DeepCore sub-array for enhanced sensitivity to low-energy events, and the IceTop array situated on the surface, designed for the detection of cosmic rays [AAA⁺17].

2.3.1 IceCube – In-Ice Array

To capture Cherenkov radiation produced by charged particles, a total of 5,160 Digital Optical Modules (DOMs) were installed at depths ranging from 1,450 to 2,450 meters beneath the ice surface, arranged on 86 vertical cables. Each cable, or string, holds 60 DOMs connected by twisted copper wire pairs. The in-ice detector array comprises 78 such strings, with DOMs spaced 17 meters apart vertically. These strings are arranged in a hexagonal pattern following a triangular grid, with about 125 meters separating them horizontally, effectively instrumenting a cubic kilometer of Antarctic ice (see Figure 2.3).

The IceCube neutrino detector produces two main event signatures, each corresponding to different interaction topologies:

1. **Cascade-like events:** In a neutral current interaction, a neutrino imparts energy to an atomic nucleus within the ice, causing the nucleus to break apart, while the neutrino itself continues on without being directly detected. This produces high-energy particles that then hadronize and decay into many secondary particles. The Cherenkov light is produced from all the particles. Since the energy loss occurs over distances smaller than the spacing between DOMs, individual particles cannot be distinguished. Instead, the detector observes a nearly spherical light pattern, referred to as a *cascade* 2.2. Cascades produce compact, localized bursts of light, enabling a fairly precise measurement of the total light output and thus the energy of the incoming neutrino.
2. **Track-like events:** These events occur when a high-energy muon neutrino undergoes a charged current interaction with a nucleus. The interaction generates a hadronic shower at the vertex and produces an outgoing muon that traverses a considerable distance through the ice, emitting Cherenkov radiation in a cone-shaped pattern along its track 2.2. Due to their directional nature, these events provide superior angular resolution, approximately 0.6 degrees [VELW21].

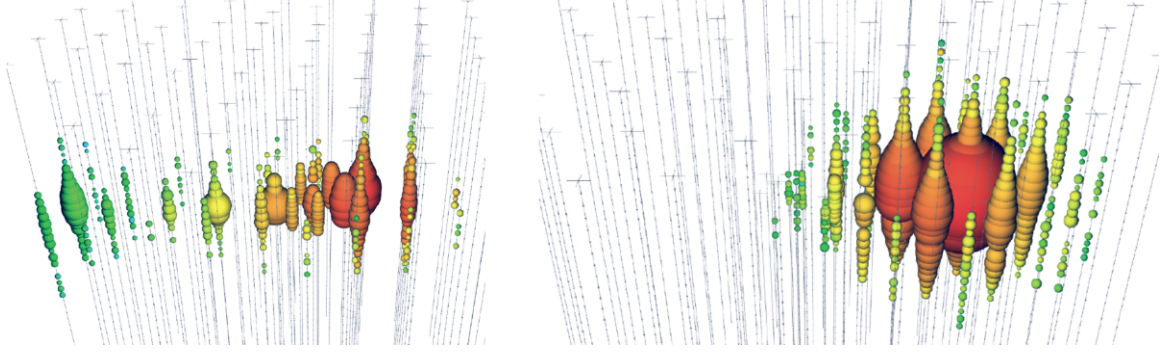


Figure 2.2: Left plot is an example of a track-like event, and the right plot is an example of a cascade-like event. Image source: [AHPdlH18]

2.3.2 DeepCore

AA portion of the in-ice DOMs is installed at depths exceeding 1750 m, creating a more densely instrumented region called *DeepCore*. This sub-array is engineered to achieve a reduced energy threshold compared to the main IceCube array, allowing it to detect neutrinos with energies down to a few GeV.

DeepCore consists of eight specialized strings of optical sensors, strategically arranged near the center of the IceCube array. These are complemented by seven adjacent standard IceCube strings. The spacing between DeepCore strings varies from 51 m to 105 m, with an average of approximately 72 m. Each DeepCore string has 50 Digital Optical Modules (DOMs) deployed at 7 m intervals between depths of 2100 m and 2450 m. Additionally, ten DOMs are placed at shallower depths above 2000 m, spaced 10 m apart. This upper segment functions as a veto region, enhancing the ability to distinguish and filter out downward-going atmospheric muons.

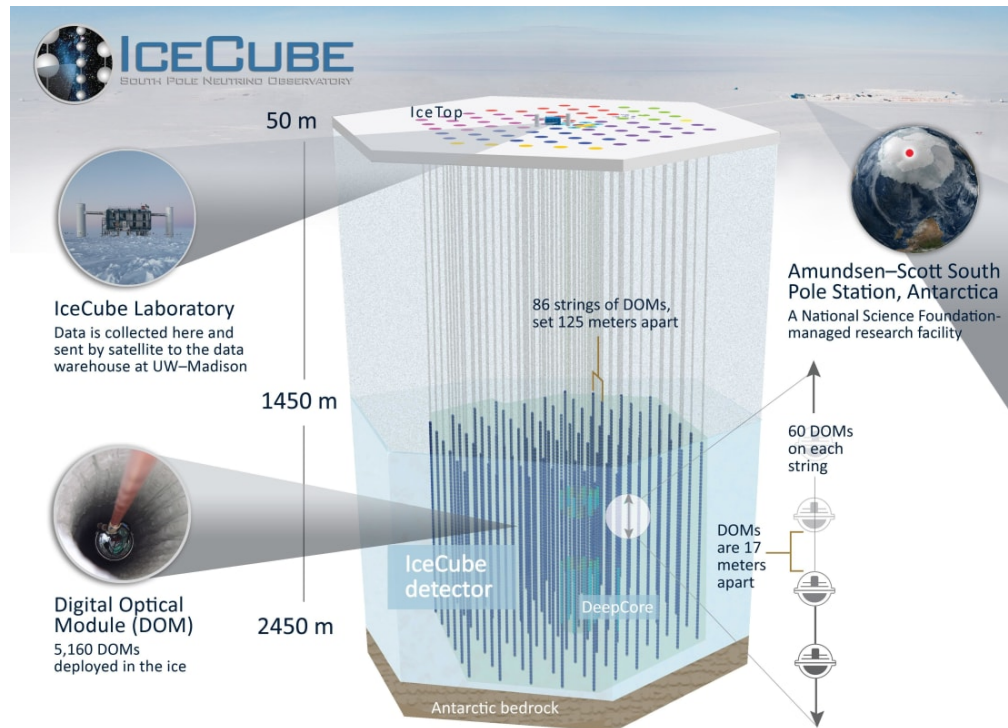


Figure 2.3: Overview of IceCube Observatory, image source: <https://icecube.wisc.edu/science/icecube/>

2.3.3 Digital Optical Modules

The Digital Optical Module (DOM) serves as the basic sensing unit within the IceCube detector. Each DOM contains a 10-inch photomultiplier tube (PMT) paired with an electronic circuit board that controls the PMT and processes its data signals. The electronics enable the PMT pulses to be recorded with nanosecond precision. Both the PMT and its circuit board are enclosed within a glass sphere that is half an inch thick. This sphere is designed to endure pressures up to 690 bar, which occur as the water refreezes following the deployment of the string. An optical gel is applied between the PMT and the glass sphere to enhance optical coupling and offer mechanical stability. DOMs are connected to the surface system via strings, which supply power and transmit data along with the GPS clock signal for precise timing.

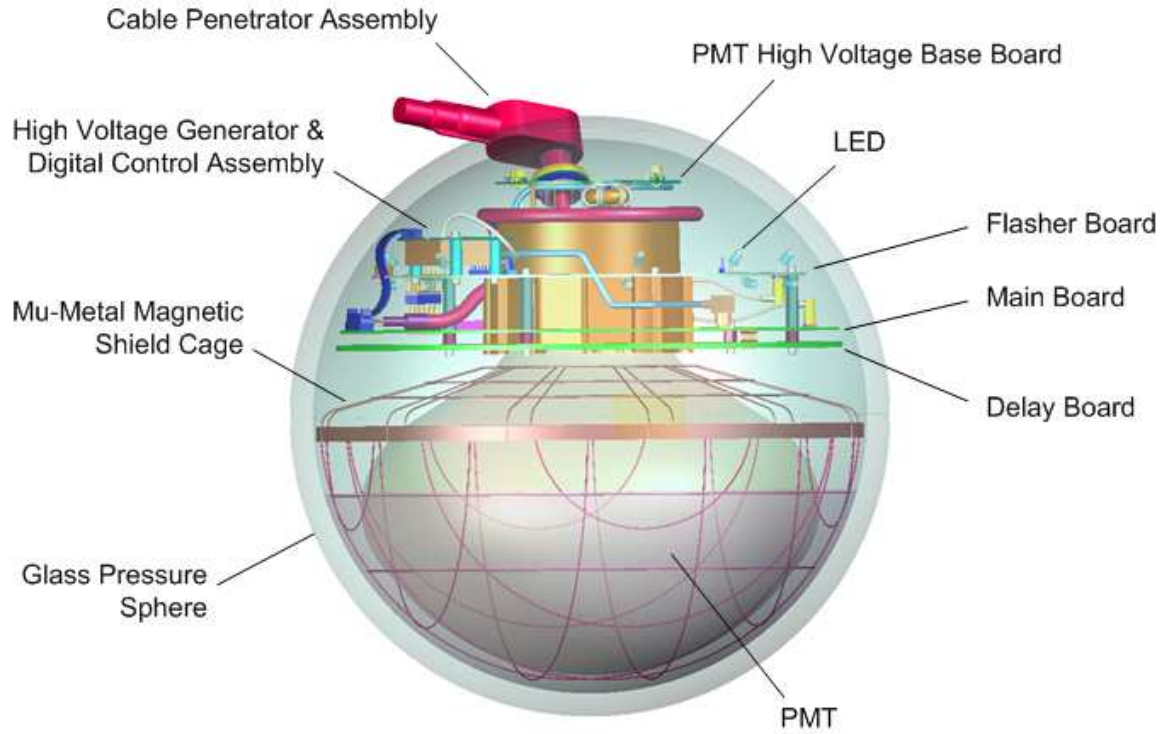


Figure 2.4: Digital Optical Module

2.3.4 IceTop

IceTop is the surface array of the IceCube Neutrino Observatory, situated at an altitude of 2835 m above sea level. It comprises 162 ice-filled tanks organized into 81 stations, aligned with the layout of the in-ice detector array. Every station contains two tanks spaced 10 m apart. Each tank is equipped with a pair of DOMs—one with high gain and the other with low gain—to capture Cherenkov light produced by particles in air showers. A denser infill region in the center corresponds to DeepCore, lowering the energy threshold to 100 TeV. IceTop measures cosmic ray air showers across energies ranging from PeV to EeV, achieving an energy resolution that improves from about 25% at 2PeV to roughly 12% for energies exceeding 10PeV.

2.3.5 IceCube Laboratory

Located at the heart of the array, the IceCube Lab contains the data acquisition and filtering equipment, which is powered by generators from the South Pole Station. The facility is kept at a stable temperature of 18°C, is shielded against electromagnetic

interference, and adheres to electrostatic safety measures because of the low humidity conditions.



Figure 2.5: image source:<https://icecube.wisc.edu/science/icecube/>

2.4 How does IceCube work

Neutrinos themselves cannot be detected directly; instead, their interactions with the ice produce charged secondary particles. These particles move through the ice at speeds exceeding that of light in the same medium, generating Cherenkov radiation. The IceCube detectors capture this emitted light, which is then digitized and assigned precise time stamps. This data is transmitted to the surface-based IceCube laboratory, where it is processed to reconstruct light patterns, enabling the determination of the incoming neutrinos' directions and energies.

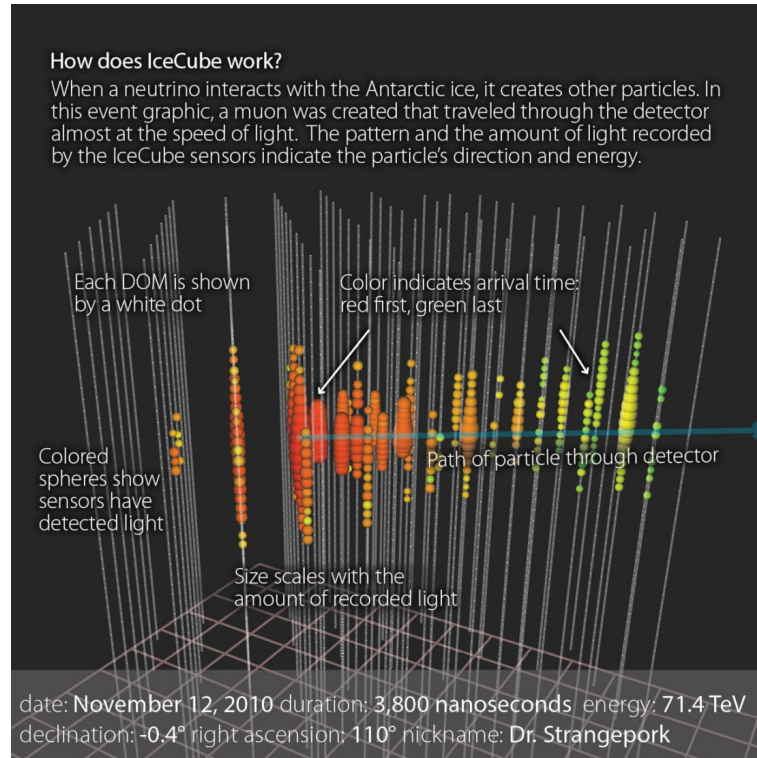


Figure 2.6: image source:<https://icecube.wisc.edu/science/icecube/>

2.5 Datas from icecube

IceCube began collecting data even while still under construction, with only a limited number of strings active. The collaboration uses the label ICXX to identify datasets, where "XX" represents the number of strings involved during data acquisition. Starting from 2011, the datasets have been named IC86-I, IC86-II, etc., indicating the consecutive years of operation with the detector fully deployed [Ice21].

IceCube describes directions within its detector using a spherical coordinate system defined by two angles: azimuth ϕ and zenith θ . The azimuth angle ϕ ranges from 0 to 2π , while the zenith angle θ spans from $\pi/2$ to π . A zenith angle of $\theta = 0$ corresponds to a direction pointing toward the Earth's center. Since the θ -axis aligns with the Earth's rotation axis, the detector effectively rotates around this axis once every 24 hours. As a result, celestial objects maintain a nearly constant zenith angle, but their azimuth angle changes continuously. Consequently, any variation in absorption due to the Earth or atmosphere depends solely on the zenith angle.

Season	Start	End	Duration	Number of Events
IC40	2008/04/04	2009/05/20	376.4 days	36900
IC59	2009/05/20	2009/05/31	352.6 days	107011
IC79	2010/06/01	2010/06/13	316.0 days	93133
IC86-I	2011/05/13	2011/05/15	332.9 days	136244
IC86-II	2011/05/26	2013/05/02	332.0 days	112858
IC86-III	2013/05/04	2013/05/18	370.7 days	127045
IC86-IV	2014/04/10	2014/05/18	357.3 days	123657
IC86-V	2015/04/24	2016/05/06	365.4 days	129311
IC86-VI	2016/05/07	2017/07/08	410.6 days	145750

Table 2.1: Summary of IceCube operational seasons, listing the start and end dates, total observation time, and recorded event counts.

2.6 IceCube Upgrade

This project plans to install 750 state-of-the-art photodetectors along with calibration equipment within the current IceCube setup [Ish19]. The upgraded instrumentation aims to enhance our knowledge of the propagation of light generated by neutrino interactions as it moves through the ice and the detector.

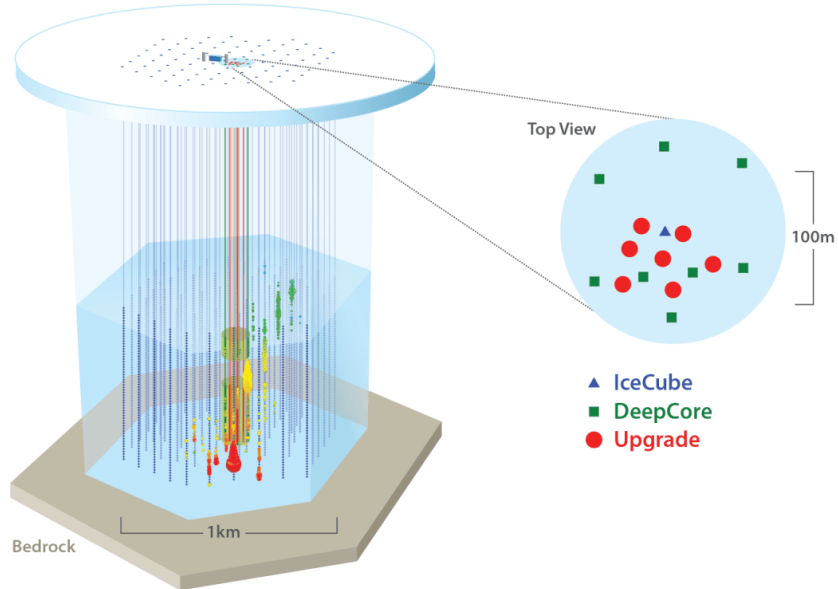


Figure 2.7: IceCube Upgrade, image source: <https://icecube.wisc.edu/science/icecube/>

2.7 IceCube Gen 2

This planned experiment at the South Pole includes three main components: a high-energy array embedded in the ice, a surface array for detecting air showers with enhanced veto functions, and a densely instrumented low-energy in-ice subarray [AAA+21b]. The high-energy section will have wider spacing between strings, enabling it to cover an effective volume up to ten times larger than that of IceCube. Its primary scientific goal is to identify and study astrophysical neutrino sources at energies around the PeV scale.

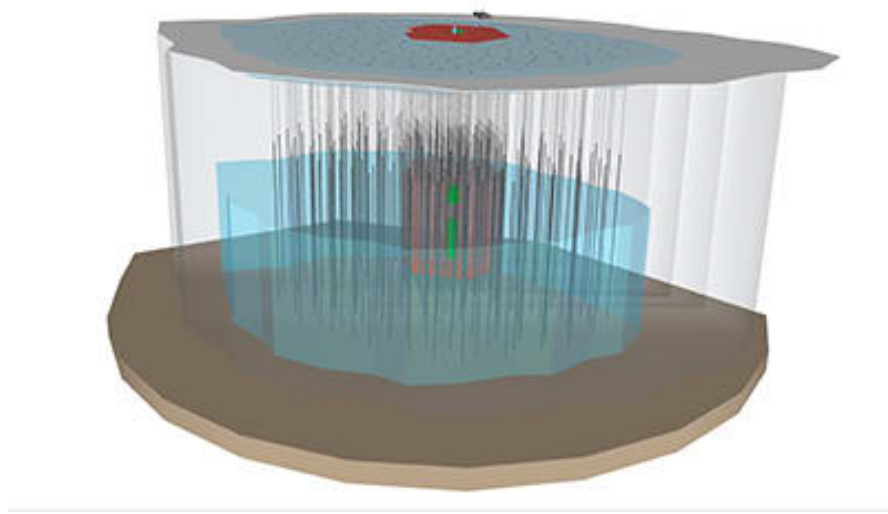


Figure 2.8: IceCube Gen 2, image source: <https://icecube.wisc.edu/science/icecube/>

Chapter 3

Statistical Methods Used in Neutrino Source Search

This chapter outlines the statistical methods employed in the search for neutrinos originating from galactic sources. First, it introduces the likelihood method, and then the hypothesis test used for analysis.

3.1 Likelihood Function

In a statistical experiment, the outcome is represented by a set of observed data points $x_1, x_2, x_3, \dots, x_n$. If these observations are assumed to be drawn from a probability density function (PDF) $P(x|\theta)$, where θ denotes an unknown parameter of the distribution, the likelihood function is formally defined as [Ros09]:

$$\mathcal{L}(x_1, x_2, \dots, x_N|\theta) = P(x_1|\theta)P(x_2|\theta) \cdots P(x_N|\theta) \quad (3.1)$$

$$= \prod_{i=1}^N P(x_i|\theta) \quad (3.2)$$

Here, θ represents the parameter value that maximizes the probability of obtaining the observed data. In other words, the maximum likelihood estimate $\hat{\theta}$ is the specific value of the parameter θ that maximizes the likelihood function $\mathcal{L}(x_1, \dots, x_n|\theta)$, given the observed dataset $x_1, x_2, \dots, x_n$.

Rather than treating the likelihood function $\mathcal{L}$ as a function of the data for a fixed parameter θ , it is more common to regard $\mathcal{L}(\theta | x_1, \dots, x_n)$ as a function of the pa-

parameter θ for a given dataset $\{x_1, \dots, x_n\}$. The parameter θ is then estimated by identifying the value that maximizes this likelihood function.

$$\frac{d\mathcal{L}(\theta)}{d\theta} = 0 \quad (3.3)$$

3.2 Unbinned Likelihood Ratio Method

Astrophysical neutrinos originating from highly energetic galactic and extragalactic sources are being observed by large-scale neutrino observatories, including IceCube and Super-Kamiokande. Various analytical techniques have been developed to identify high-energy, point-like neutrino sources using deep-embedded detectors in polar ice or underwater environments. Here we are discussing the unbinned likelihood ratio method, which is better than the binned likelihood method.

In IceCube all-sky point source data, which is taken from 2008–2018, The dataset includes information on muons distributed across the sky, with each event characterized by a reconstructed arrival direction—specified in terms of right ascension and declination—as well as an associated positional uncertainty and an estimate of the muon energy.

The unbinned likelihood approach is employed to assess the existence and intensity of a possible point-like neutrino source by statistically separating it from the isotropic background of randomly distributed events [CAA+19]. .

3.2.1 Probability Density Function (PDF)

The probability density function (PDF) characterizes the likelihood of observing a specific value in a dataset. It plays a central role in likelihood estimation. It is normalized such that its total integral (or sum) equals one. Depending on whether the variable is continuous or discrete, the PDF represents a continuous distribution or a probability mass function, respectively.

In the context of a point-like neutrino source, two distinct probability density functions (PDFs) are considered: the signal PDF, which characterizes the expected distribution of events from the source, and the background PDF, which models the distribution of events arising from background noise.

For a point-source neutrino analysis [PD22], given a dataset containing N events, the probability density associated with the i^{th} event is expressed as:

$$P_i = \frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \quad (3.4)$$

Here, N represents the total number of events in the dataset, and n_s denotes the number of signal events attributed to the galactic source under investigation. The functions S_i and B_i correspond to the signal and background probability density functions (PDFs) evaluated for the i^{th} neutrino event, respectively. The signal PDF is given by:

$$S_i = \frac{1}{2\pi\sigma_i^2} \exp\left(-\frac{|\theta_i - \theta|^2}{2\sigma_i^2}\right) \quad (3.5)$$

In this expression, $|\theta_i - \theta|$ denotes the angular separation between the direction of the i^{th} neutrino and the galactic source, while σ_i represents the angular uncertainty associated with the neutrino's reconstructed position, measured in radians.

The background probability density function is derived from the solid angle covering a declination range of $\pm 5^\circ$ centered on the galactic source under observation:

$$B_i = \frac{1}{\Omega_{\delta \pm 5^\circ}} \quad (3.6)$$

Now, the point source likelihood is expressed as:

$$\mathcal{L}(n_s) = \prod_{i=1}^N P_i \quad (3.7)$$

$$= \prod_{i=1}^N \left[\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \right] \quad (3.8)$$

For numerical ease, we will use the logarithm of likelihood instead of likelihood itself, although it does not affect the position of the maximum of $\mathcal{L}$:

$$\log(\mathcal{L}(n_s)) = \sum_{i=1}^N \log\left(\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i\right) \quad (3.9)$$

3.3 Hypothesis Testing

The purpose of a hypothesis test is to evaluate the probability of obtaining the observed experimental outcome under a specified hypothesis. To quantify this evaluation, a scalar value known as the test statistic is defined:

$$\lambda = 2 \log \left(\frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(0)} \right) \quad (3.10)$$

When the maximized likelihood $\mathcal{L}(n_s)$ closely aligns with the null hypothesis H_0 , the test statistic λ approaches zero. Nonetheless, because of inherent statistical fluctuations in actual data, the distribution of λ under H_0 is not a delta function at zero, but instead exhibits a spread around this value.

P-value

It is a scalar which indicates how well the outcome of an experiment agrees with the null hypothesis H_0 [Ros09]. Given that λ_{exp} is the test statistic measured in the experiment and $H_0(\lambda)$ denotes the distribution of the test statistic under the null hypothesis, the p-value is expressed as:

$$p = 1 - \int_0^{\lambda_{\text{exp}}} H_0(\lambda) d\lambda \quad (3.11)$$

$$= \int_{\lambda_{\text{exp}}}^{\infty} H_0(\lambda) d\lambda. \quad (3.12)$$

If the p-value is smaller, it implies that the observed result is less likely to occur under the assumption that the null hypothesis H_0 is true, which means it is inconsistent with H_0 .

Hypothesis testing

In hypothesis testing, we compare two hypotheses with each other. These are the null hypothesis, which is usually denoted as H_0 , representing the known model, and the alternative hypothesis, denoted as H_1 , extends the null hypothesis H_0 by including an additional unknown component. This extra component may correspond to phenomena such as an undiscovered resonance in a spectrum or a candidate neutrino point source. In the context of this analysis, the data are described by the following hypotheses [BDDP⁺08]:

- H_0 : The observed data arise exclusively from background atmospheric neutrino events.
- H_1 : The observed data consist of background atmospheric neutrinos combined with astrophysical neutrino events originating from a source with a certain flux strength.

$$\lambda = 2 \cdot \log \left(\frac{P(\text{Data}|H_s)}{P(\text{Data}|H_0)} \right) \quad (3.13)$$

The larger value of λ indicates that the observed data are less consistent with the background-only hypothesis H_0 .

Before doing the experiment, A significance level is selected as the criterion for rejecting the null hypothesis H_0 , commonly S sigma or a predefined value. This chosen level determines the threshold value λ_{thr} for the test statistic.

If the observed test statistic λ_{exp} exceeds the threshold value λ_{thr} , the null hypothesis H_0 is rejected in favor of the alternative hypothesis. Conversely, if $\lambda_{\text{exp}} < \lambda_{\text{thr}}$, the null hypothesis H_0 is not rejected, and the alternative hypothesis is consequently rejected. Here:

	Conclude H_0 is True	Conclude H_1 is True
H_0 is True	✓	Type I Error
H_1 is True	Type II Error	✓

Table 3.1: Possible outcomes and associated errors in statistical hypothesis testing

Type I error: This implies rejection of H_0 when H_0 is true. It is called a *false positive*. Chance for Type I error is

$$\alpha = \int_{\lambda_{\text{thr}}}^{\infty} H_0(\lambda) d\lambda \quad (3.14)$$

Type II error: It means the rejection of H_1 if H_1 is true, or we can say accepting H_0 when H_1 is true, which is called a *false negative*.

$$\beta = \int_0^{\lambda_{\text{thr}}} H_1(\lambda) d\lambda \quad (3.15)$$

There is no optimal choice of λ_{thr} . In high-energy physics, it is commonly set to 5σ .

To determine whether the observed data are consistent with the background-only hypothesis, we employ a statistical approach based on the construction of a

p-value (N_σ)	Common Rule	Revised Rule
$N_\sigma = 1$	Mild	None
$N_\sigma = 2$	Significant	Mild
$N_\sigma = 3$	Highly Specific	Significant
$N_\sigma = 4$	Overwhelming	Highly Significant
$N_\sigma = 5$	—	Overwhelming

Table 3.2: Interpretation of p-value (N_σ) in terms of evidence against H_0

test statistic, denoted by λ . The method of maximum likelihood, widely used for parameter estimation in statistics, is particularly useful for this analysis. In this context, the test statistic is formulated as the ratio between the likelihood function evaluated under the alternative hypothesis H_1 using the best-fit value of the signal parameter n_s , and the likelihood function evaluated under the null hypothesis, where $n_s = 0$:

$$\lambda = 2 \log \left(\frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(0)} \right) \quad (3.16)$$

$$= 2 \log \left(\frac{\prod_{i=1}^N \left(\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \right)}{\prod_{i=1}^N B_i} \right) \quad (3.17)$$

$$= 2 \sum_{i=1}^N \log \left(\frac{n_s}{N} \cdot \frac{S_i}{B_i} + \left(1 - \frac{n_s}{N}\right) \right) \quad (3.18)$$

$$= 2 \sum_{i=1}^N \log(n_s \chi_i + 1) \quad (3.19)$$

With:

$$\chi_i = \frac{1}{N} \left(\frac{S_i}{B_i} - 1 \right) \quad (3.20)$$

The quantity χ denotes the ratio between the signal and background probability density functions, serving as an indicator of the likelihood that a given event originates from a true signal rather than background noise. Rather than maximizing the logarithm of the likelihood function directly, the optimization is performed on the test statistic itself. Since $\mathcal{L}(0)$ remains a fixed reference value, its presence does not influence the location of the maximum in the parameter space.

Assuming the background-only hypothesis holds, Wilks' theorem [Wil38] implies that the test statistic follows a chi-square distribution, where the degrees of freedom correspond to the number of free parameters being optimized. Although the parameter

n_s , representing the number of signal events, can theoretically assume both positive and negative values, negative estimates are conventionally constrained to $n_s = 0$ to prevent unphysical interpretations. Consequently, in the presence of statistical fluctuations, the test statistic is expected to be zero or greater, with approximately 50% of background-only realizations yielding a test statistic close to zero.

Chapter 4

Search for spatial coincidence between IceCube neutrinos and gamma-ray bright red dwarfs

In this chapter, we discuss the search for neutrino emission from the aforementioned 10 red dwarfs using publicly available muon track data from IceCube. The outline of this chapter is as follows: a description of the red dwarfs, the neutrino dataset used for the analysis, followed by the method of analysis and the results.

4.1 Red Dwarf as a Source of Galactic Cosmic Rays

Red dwarfs are low-mass stars situated in the lower-right region of the Hertzsprung–Russell (HR) diagram 4.1. Their masses typically range between 0.075 and 0.5 times that of the Sun, and their surface temperatures lie between 2500 and 3000 K. Despite having lower mass, these stars often exhibit intense and frequent stellar activity, primarily due to their fully convective interiors that give rise to strong magnetic fields. The occurrence frequency of flares from Sun-like M dwarf stars can be characterized by a power law:

$$\frac{dN}{dE} \propto E^\alpha, \quad (4.1)$$

with the index α varying between 1.7 and 2.4. The total energy released in a single flare from a red dwarf is about 10^{32} – 10^{35} erg, which is thousands of times higher than the energy of a solar flare.

It is known that red dwarfs are the most numerous stars, accounting for more than

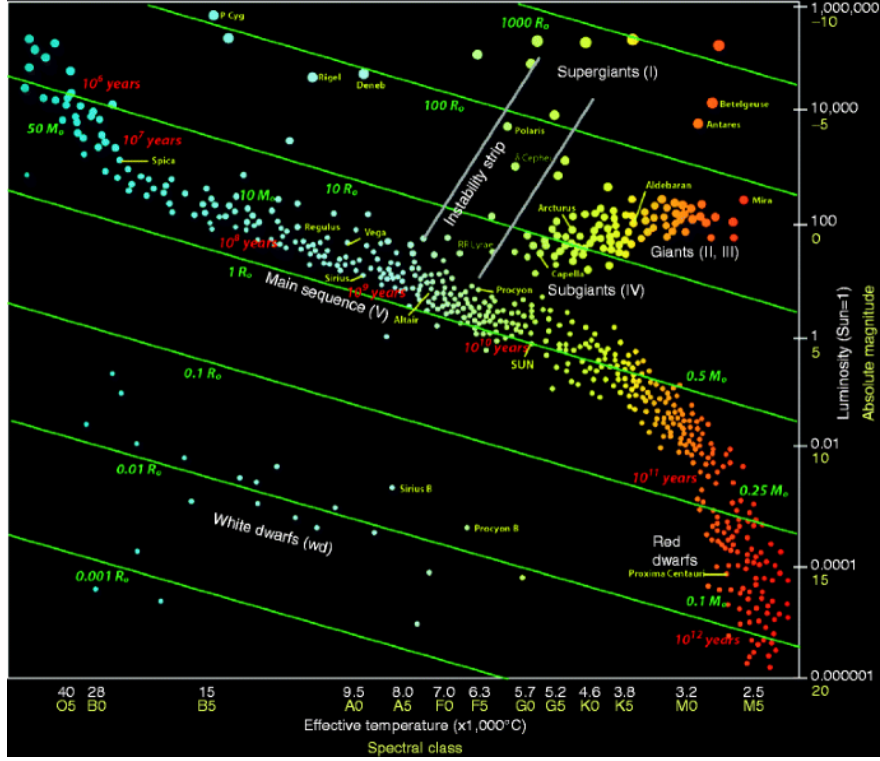


Figure 4.1: Hertzsprung–Russell (HR) diagram showing the position of red dwarfs at the bottom right, characterized by low luminosity and low surface temperature.

70% of the galactic stellar population. The estimated total energy of cosmic rays produced in the stellar flares of galactic dwarf stars is about 10^{51} – 10^{53} ergs, which is comparable to the estimated energy budget of cosmic rays in the galactic disk. Therefore, it is conjectured that red dwarfs may be among the sources of galactic cosmic rays with energies up to 10^{14} eV.

Energetic outbursts have been detected from red dwarfs at high energies—from hard X-rays to TeV gamma-rays. Most recently, the SHALON TeV gamma-ray telescope detected very high-energy gamma-ray emission between 800 GeV and 20 TeV from eight red dwarfs: V388 Cas, V547 Cas, V780 Tau, V962 Tau, V1589 Cyg, GJ 1078, GJ 3684, and GL 851.1. Subsequently, these red dwarfs have been proposed as sources of ultra-high-energy cosmic rays.

However, no corresponding gamma-ray emission was seen at GeV or MeV energies. Song and Paglione reported a 5σ detection of pulsed GeV γ -ray emission from another M dwarf, TVLM 513-46546, with Fermi-LAT. A very hard X-ray outburst was also detected by the SWIFT-BAT telescope from another M dwarf, DG CVn, in the 0.3–

100 keV range in April 2014, although Fermi-LAT observed no coincident gamma-ray emission.

Although there are no current estimates for high-energy neutrino flux emission from red dwarfs, these stars could generate cosmic rays with energies up to 10^{14} eV during their flaring states. The TeV gamma rays detected from these red dwarfs indicate that they produce flares with total energies of 10^{32} – 10^{35} ergs.

Considering the statistics of M dwarfs in our galaxy and assuming a flare frequency of around 36 per year for about 10% of the observed red dwarfs, the total energy released from these flares matches the total energy of cosmic rays in the galactic disk, estimated to be $\sim 10^{54}$ ergs.

During these flares, the accelerated protons with energies up to 10^{14} eV undergo spallation with the surrounding atmosphere outside the star. Similar mechanisms have also been proposed for the Sun. These interactions produce pions, and consequently, one would also expect neutrino emission from the aforementioned hadronic interactions, along with gamma-rays produced from pion decay.

Therefore, these considerations provide enough motivation to search for neutrino emissions to complement recent searches for MeV and GeV gamma-rays from these red dwarfs.

4.2 Dataset

The neutrinos used in this work are taken from the IceCube 10 year muon track data. IceCube is a detector located at the South Pole, which detects neutrinos through the Cherenkov radiation emitted by the charged leptons traveling through the ice.

This dataset consists of 1,134,431 track-like neutrino candidates, which were collected between April 2008 (IC-40) and July 2018 (IC 86-V11) from four different phases of the experiment, each having different livetime. Note that for our analysis, we have used the augmented dataset analyzed in [ZB22]¹, which corrects for 38 duplicate events found in the IceCube public muon track data due to angular reconstruction

¹This dataset is available at https://github.com/beizhouphys/IceCube_data_2008--2018_double_counting_corrected

error [ZB22]. For each neutrino, the dataset consists of right ascension (RA), declination (δ), error in the position, and reconstructed muon energy.

4.3 Analysis and Results

For our analysis, we are using the unbinned maximum likelihood ratio method (See [PD22] and references therein, and first proposed in [BDD⁺08b]). We use neutrinos within a declination of $\pm 5^\circ$ of the red dwarf. For a dataset containing N events, if n_s is the number of signals attributed to the red dwarf, then the likelihood of the entire dataset is given by:

$$\mathcal{L}(n_s) = \prod_{i=1}^N \left[\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \right] \quad (4.2)$$

where S_i is the signal PDF and B_i is the background PDF. Here the signal PDF is given by:

$$S_i = \frac{1}{2\pi\sigma_i^2} e^{-(|\theta_i - \theta_s|)^2 / 2\sigma_i^2} \quad (4.3)$$

where $|\theta_i - \theta_s|$ is the angular distance between the neutrino and the red dwarf, σ_i is the angular uncertainty in the neutrino position, in radians. The background PDF can be obtained from the solid angle within the declination (δ) of $\pm 5^\circ$ around each red dwarf and is given as follows [BDD⁺08b]:

$$B_i = \frac{1}{\Omega_{\delta \pm 5^\circ}} \quad (4.4)$$

We then evaluate $\mathcal{L}(n_s)$ for different values of n_s , and designate the value of n_s , for which this likelihood is maximized as $\hat{n}_s$. Now to ascertain the significance of the signal, we define the test statistics (TS) based on the likelihood ratio, with respect to the null hypothesis in the absence of a signal, ($\mathcal{L}(0)$) as follows:

$$TS = 2 \log \frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(0)}. \quad (4.5)$$

For the null hypothesis, TS behaves like a χ^2 distribution for one degree of freedom [Wil38]. The detection significance statistic can be obtained from $\sqrt{TS}$. This statistics is also used in gamma-ray astronomy (eg. [MD24]). For a statistically significant detection of $> 5\sigma$, $TS > 25$. For each of our red dwarfs, we determine $\hat{n}_s$ and evaluate TS according to Eq 6.4. These values of TS , along with the values of $\hat{n}_s$ can be found in Table 5.1. As we can see, none of the red dwarfs show a TS value

> 25 . The largest TS value is seen for GJ 1078 with TS of 2.38, whose significance is less than 2σ . Therefore, the observed signal events are consistent with background, and there is no evidence for any spatial association between red dwarfs and IceCube neutrinos, which are observed as muon tracks. Consequently, we calculate the 95% c.l. upper flux limits by calculating the value of n_s for which $\Delta(TS) = -3.84$ [Z⁺20], where $\Delta TS = TS - TS_{max}$.

Red dwarf	RA (°)	$\delta(^{\circ})$	$\hat{n}_s$	TS	Upper limit
V388 Cas	15.8326	+62.365	0.0	0.0	16.2
V547 Cas	8.1229	+67.235	0.0	0.0	16.7
V780 Tau	85.107	+24.802	6.00	0.35	29.8
V962 Tau	86.466	+22.879	0.0	0.0	13.8
V1589 Cyg	310.705	+41.38	0.0	0.0	12.7
GJ 1078	80.95	+22.54	20.39	2.38	51.2
GJ 3684	176.77	+70.03	0.0	0.0	14.9
GL 851.1	333.0267	+31.56	0.0	0.0	11.1
TVLM 513-46546	225.284	22.834	0.0	0.0	20.8
DG CVn	202.944	29.28	2.42	0.05	27.2

Table 4.1: Results from spatial coincidence analysis between IceCube neutrinos and red dwarfs. None of the red dwarfs show a detection significance $> 5\sigma$, corresponding to $TS > 25$. Therefore, we do not find any spatial association between the red dwarfs and IceCube neutrinos. The last column shows the 95% c.l. upper limit on the number of signal neutrino events.

Chapter 5

Search for spatial coincidence between magnetars and IceCube detected neutrinos

5.1 Magnetars

Magnetars represent a unique subclass of neutron stars, notable for their extreme variability observed mainly in the X-ray and soft gamma-ray bands. These variations can range from brief millisecond-scale flashes to prolonged outbursts lasting several weeks. The source of their emission is attributed to the dissipation of their ultra-strong internal magnetic fields. The term *magnetar* was introduced in 1992 by Duncan and Thompson to reflect this distinctive magnetic origin.

They exhibit extremely strong magnetic fields, with values up to 10^{15} G, and X-ray luminosities between 10^{31} – 10^{34} erg/s. The X-ray emission of magnetars originates from the decay and instability of their ultra-strong magnetic fields. Observationally, they are manifested as Soft Gamma-Ray Repeaters (SGRs) and Anomalous X-ray Pulsars (AXPs).

Several theoretical models for neutrino emission from magnetars have been proposed, applicable during both their quiescent phases and their outbursts. Although historically most known magnetars were galactic, we have now detected extragalactic magnetar flares from several sources. Magnetars show different types of transient emission, such as:

- **Short bursts:** Duration $\sim 0.1\text{--}0.2$ seconds, peak luminosity $10^{38}\text{--}10^{40}$ erg.
- **Intermediate flares:** Duration from several seconds to years, luminosity $10^{41}\text{--}10^{42}$ erg.
- **Giant flares:** Most energetic, luminosity $10^{43}\text{--}10^{45}$ erg/s.

However, these giant flares from galactic magnetars were observed from only three sources before the start of IceCube operations. From 2008 to 2018, the period for which public IceCube data is available, no magnetars showed large transient flares compared to quiescent emission. Moreover, the exact timing of outbursts is often difficult to determine. Observations suggest that quiescent luminosity in many SGRs remains elevated for long periods post-flaring, which may imply increased neutrino flux during these post-burst phases.

The first-ever search for neutrinos from magnetars using a 1000 m^2 detector was performed by Super-Kamiokande, targeting neutrinos coincident with four SGR bursts using upward-going muon samples. Two neutrinos were observed in spatial coincidence within $\pm 5^\circ$ and within one day post-outburst, but the p -value was consistent with background after accounting for the look-elsewhere effect. Subsequent searches during both flaring and quiescent phases using Super-K's highest energy neutrino subset yielded no detection. A proof-of-principle study with 14 years of IceCube data has also been conducted to evaluate sensitivity to galactic magnetars.

5.1.1 Neutrino Production Mechanism in Magnetars

Magnetars exhibit surface magnetic fields of the order $10^{13}\text{--}10^{15}$ G and emit strong X-ray and thermal radiation early in their lifetimes due to magnetic field decay, maintaining high X-ray luminosity over extended periods.

Young magnetars with oppositely oriented magnetic field and spin moment can emit high-energy neutrinos from their polar caps, as they can accelerate cosmic rays to very high energies. Neutrino emission from high-energy photons occurs via pion decay through photomeson interactions. These pions are produced when protons interact with ambient photons or matter. Pulsars, acting as unipolar generators, induce a large potential to accelerate protons. These protons, upon collision with target photons or matter, can produce neutrinos via pion decay.

One major challenge for neutrino production in pulsar environments is the insufficient column density of targets for pion production. Therefore, neutrino production is typically expected in the pulsar wind nebula. However, it has been shown that the conditions for photomeson interactions can occur even within the magnetospheres of magnetars (pulsars with $B_s \geq 10^{15}$ G), provided they are young enough to accelerate protons beyond the photomeson threshold.

The two primary energy sources in a magnetar are:

- **Spin-down power:** Represents the loss of rotational energy, used for accelerating protons.
- **Magnetic power:** Originates from the decaying magnetic field and provides a dense photon field near the surface.

Together, these facilitate photomeson interactions. In slowly rotating magnetars (such as SGRs and AXPs), magnetic power dominates over spin-down power. However, in their early life stages, rapid rotation may allow spin-down power to exceed magnetic power. The dominant process for neutrino production is the Δ -resonance in photon-meson interaction:

$$p\gamma \rightarrow \Delta \rightarrow n\pi^+ \rightarrow n\nu_\mu\mu^+ \rightarrow n\nu_\mu e^+ \nu_e \bar{\nu}_\mu \quad (5.1)$$

SGRs, after flaring, exhibit enhanced quiescent luminosity for extended periods, which may contribute to a higher neutrino flux during the post-burst phase.

5.1.2 Dataset

For this analysis, we have used neutrinos from the IceCube public 10-year muon track data [A⁺21]. This data set consists of 1,134,431 neutrinos, which were collected between April 2008 (IC-40) and July 2018 (IC86-VII) from different phases of the experiment, each having a different lifetime. For each neutrino, the data set consists of right ascension (RA), declination, uncertainty in the direction of the track, and reconstructed muon energy. Note that for our analysis, we have used the augmented data set analyzed in [ZB22], which removes some duplicates from the IceCube dataset ¹. The list of galactic magnetars used for this work has been obtained from

¹This dataset is available at https://github.com/beizhouphys/IceCube_data_2008--2018_double_counting_corrected

the McGill Online Magnetar Catalog [OK14],² and includes 31 magnetars, of which 16 are SGRs (12 confirmed candidates), and 14 are AXPs (12 confirmed candidates). One of the magnetar candidates includes PSR J1846-0258, which has been classified as a young, rotation-powered pulsar, but has also undergone a magnetar-like outburst in 2006 [GGG⁺08]. In addition to these galactic magnetars, we also considered six extragalactic magnetars, for which flares were detected in high-energy gamma-rays. These include GRB 200415A [FAA⁺21], GRB 231115A [TSv⁺24], GRB 051103, GRB 070201, GRB 070222, and GRB 180128A.

5.1.3 Analysis

For our analysis, we use the unbinned maximum likelihood ratio method [BDD⁺08b]. We consider neutrinos within a declination of $\pm 5^\circ$ of the magnetars. For a dataset containing N events, if n_s is the total number of signal events attributed to the magnetars, then the likelihood is given by:

$$\mathcal{L}(n_s) = \prod_{i=1}^N \left[\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \right], \quad (5.2)$$

where S_i is the signal probability density function (PDF) and B_i is the background PDF. Here the signal PDF is given by:

$$S_i = \frac{1}{2\pi\sigma_i^2} e^{-(|\theta_i - \theta_j|)^2 / 2\sigma_i^2}, \quad (5.3)$$

where $|\theta_i - \theta_j|$ is the angular distance between the neutrino and the magnetar (indexed by j); σ_i is the angular uncertainty in the neutrino position expressed in radians. Similarly to other works which have analyzed the IceCube public data [ZKL21, PD22, LZL22], we have estimated the background PDF from the data. We assume that the background neutrinos are uniformly distributed across the region of interest and do not have any dependence on the right ascension. The background PDF can therefore be obtained from the solid angle within the declination band (δ) of $\pm 5^\circ$ around each magnetar, and is given as follows [PD22, LZL22, BDD⁺08b]:

$$B_i = \frac{1}{\Omega_{\delta \pm 5^\circ}} \quad (5.4)$$

²This dataset is available at <http://www.physics.mcgill.ca/~pulsar/magnetar/main.html>

Note that the standard IceCube analysis also uses an additional PDF for the signal and background, which takes into account the neutrino energy [A⁺17]. However, the public IceCube muon dataset only contains the muon energy and it is not trivial to reconstruct the neutrino energy-based PDF using the smearing matrix provided along with the public data. Furthermore, there is no theoretical guidance on the expected neutrino emission spectral energy distribution from a magnetar. Therefore, the energy-based PDF is usually ignored in most other point source searches with the public IceCube data [ZKL21, HLV19, PD22, BPDF24, LZL22, LZ20, LZ24]. We do not consider this in our analysis.

Now to ascertain the significance of the signal, we define the test statistics (TS) as follows:

$$TS(n_s) = 2 \log \frac{\mathcal{L}(n_s)}{\mathcal{L}(0)} \quad (5.5)$$

The best-fit value of the number of signal events (n_s) is obtained by maximizing TS . For the null hypothesis, $TS(n_s)$ behaves like a χ^2 distribution asymptotically for one degree of freedom [Wil38]. Note however that the constraint that the signal strength is non-negative and the limited size of the dataset may cause deviations from the expected distribution based on Wilks theorem. However, the distribution of TS for the null hypothesis has been evaluated using random source locations in [LZL22], and has been shown to agree with that expected from Wilk’s theorem. We shall also independently check the distribution of TS for the null hypothesis in Section 5.1.4. Therefore, the significance of the detection can be approximated by $\sqrt{TS}$. For a statistically significant detection corresponding to $> 5\sigma$ detection, TS must be > 25 .

5.1.4 Results

For each of our magnetars, we calculate the best fit n_s that maximizes TS, according to Eq. 5.5. These TS values, along with the best-fit value of n_s , can be found in Table 5.1. As we can see, none of the magnetars show a TS value > 25 . The largest TS value is seen for 3XMM J185246.6+003317, with TS of 5.34, which corresponds to a significance of around 2.31σ (based on Wilks’ theorem).

To independently evaluate the p -value for the magnetar with the highest TS value, we plot the distribution of TS for the null hypothesis (of no signal). For this purpose, similar to [ZKL21], we evaluated TS at 1000 locations on the celestial sphere with RA uniformly distributed between 0 and 360° and $\sin(\delta)$ uniformly distributed between -1

and 1. We then evaluate TS for each of these random locations in the same way as for each of these magnetars. The distribution of TS for each of these random locations can be found in Fig. 5.1 and constitutes the null hypothesis. The p -value can then be found by calculating the ratio of number of simulated locations with TS values greater than that observed for a given magnetar to the total number of simulated TS values. If we consider the maximum TS value of 5.34 (for XMM J185246.6 + 003317), we find a total of 9 simulated locations with $TS > 5.34$, which corresponds to a p -value of 0.009, which in turn corresponds to a significance of 2.36σ [GD17], which roughly agrees with $\sqrt{TS}$, which we estimated earlier. If one considers the look-elsewhere effect, incorporating the total number of signal events looked at, the significance will be reduced even further.

Therefore, the observed signal events are consistent with background and there is no evidence for any spatial association between our catalog of magnetars and IceCube neutrinos, which are observed as muon tracks. Therefore, none of the known magnetars contribute towards the diffuse IceCube neutrino signal. Consequently, we calculate the 95% c.l. upper limits on the observed signal events by calculating the value of n_s for which $TS - TS_{max} = -3.84$ [Z⁺20]. In order to compare to theoretical models, we first obtain the differential neutrino flux limits. In order to do this, we assume that the magnetar neutrino spectrum can be characterized as a function of neutrino energy (E_ν) by a power law with spectral index Γ .

$$\Phi_\nu(E_\nu) = \phi_0 (E_\nu/100 \text{ TeV})^\Gamma \quad (5.6)$$

where ϕ_0 is the flux normalization at 100 TeV. For this purpose we calculate the effective acceptance across all seasons for a magnetar with declination (δ_j) which is given by

$$A_{acc} = \sum_i^k T_k \times \int A_{eff}(E_\nu, \delta_j) \Phi_\nu(E_\nu) dE_\nu, \quad (5.7)$$

where $\phi_\nu(E_\nu)$ is defined in Eq. 5.6. To evaluate the acceptance, we have assumed the spectral index ($\Gamma = -2.53$), similar to [PD24]. In Eq 5.7, T_k is the livetime of each season and the sum is over all the IceCube seasons. The IceCube effective area ($A_{eff}(E_\nu, \delta_j)$) is a function of the neutrino energy in addition to the δ_j . For each magnetar, we then calculated the neutrino flux limits at $E_\nu = 100$ TeV by dividing the 95% upper limit on the number of signal events by A_{acc} . These flux limits can be trivially computed for any other neutrino energy using Eq. 5.6. These neutrino flux

limits can be found in the last column in Table 5.1.

The corresponding fluxes for neutrino induced muons produced from magnetars where spin and magnetic moments point in opposite direction has been estimated for four objects, and found to be $\mathcal{O}(0.01-10) \text{ km}^{-2}\text{yr}^{-1}$ [ZDM⁺03] with the maximum expected flux for SGR 1900+14, equal to $13 \text{ km}^{-2}\text{yr}^{-1}$. The corresponding fluxes of PeV energy neutrino-induced muons produced from acceleration in the magnetic polar caps are even lower, viz. $2.5 \times 10^{-4} \text{ km}^{-2}\text{yr}^{-1}$ [DRD16].

In order to compare our neutrino flux limit to theoretical predictions, we convert our estimated neutrino flux limits to upward muon flux limits (Φ_μ) using the same prescription as in [DAH⁺08]:

$$\Phi_\mu = \int_{E_{th}}^{\infty} \Phi_\nu(E_\nu) P(\nu \rightarrow \mu) dE_\nu, \quad (5.8)$$

where $P(\nu \rightarrow \mu)$ is the muon to neutrino conversion probability obtained from $P(\nu \rightarrow \mu) \approx 1.3 \times 10^{-6} (E_\nu/\text{TeV})$; E_{th} is the muon energy threshold which is assumed to be 200 GeV. Using this we obtain the 95% c.l. muon flux limits for SGR 1900+14, SGR 0526-66, 1E 1048-66, and SGR 1806-20 to be $\sim 10^6 \text{ km}^{-2}\text{yr}^{-1}$. Therefore, our muon flux limits are not stringent enough to constrain models of neutrino emission from rotating magnetars with spin and magnetic moments oppositely aligned [ZDM⁺03] or the mechanism proposed in [DRD16]. For a more robust test from individual magnetars., next-generation detectors such as IceCube-Gen2 [AAA⁺21a] would be needed.

5.2 Stacked Analysis

We now perform a stacked search analysis for the magnetars, following the same methodology as in [PD24]. We quickly summarize the procedure followed (using the same notation as [PD24]), and highlight some differences with respect to single-source analysis.

The background PDF (B_i) for stacked analysis is determined by the solid angle within 5° declination band around each event i ($\Omega_{\delta_i \pm 5^\circ}$), and scale by the fraction of events

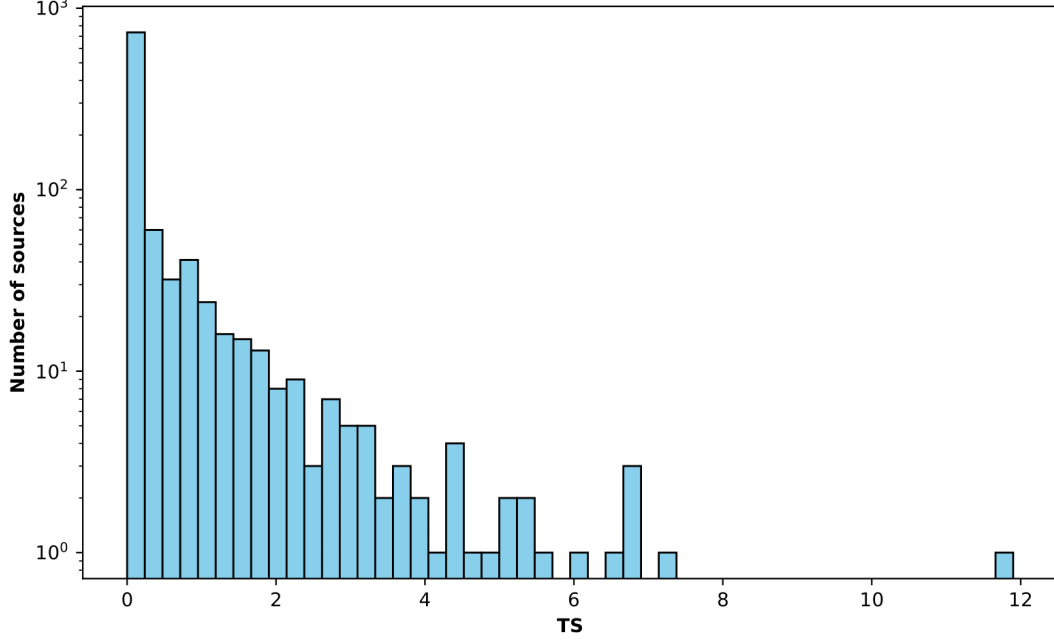


Figure 5.1: Distribution of the test statistic (TS) under the null hypothesis using 1000 randomly selected sky positions. Right Ascension (RA) values are uniformly distributed between 0° and 360° , and Declination is chosen such that $\sin(\delta)$ is uniformly distributed between -1 and 1 .

within this declination band:

$$B_i = \frac{\mathcal{N}_i}{N\Omega_{\delta_i \pm 5^\circ}}, \quad (5.9)$$

where $\mathcal{N}_i$ is the number of events within the $\pm 5^\circ$ declination band of event i and N is the total number of neutrinos. We also plot this background PDF as a function of each season used in the analysis in Fig. 5.2. The signal PDF (S_i) is a weighted average of the signal PDFs (S_{ij}) of all the sources:

$$S_i = \frac{\sum_j \omega_{acc,j} \omega_{model,j} S_{ij}}{\sum_j \omega_{acc,j} \omega_{model,j}}, \quad (5.10)$$

where the sum j extends over all the 37 magnetars in the catalog, S_{ij} is given by the same equation as in Eq 6.2. In Eq 5.10, $\omega_{model,j}$ corresponds to the signal weight used for a given magnetar. Since there is not much guidance from theoretical models, we choose $\omega_{model,j} = 1$ for our analysis. However, it is straightforward to extend this analysis to other signal weighting factors similar to [PD24]. Finally, $\omega_{acc,j}$ is defined

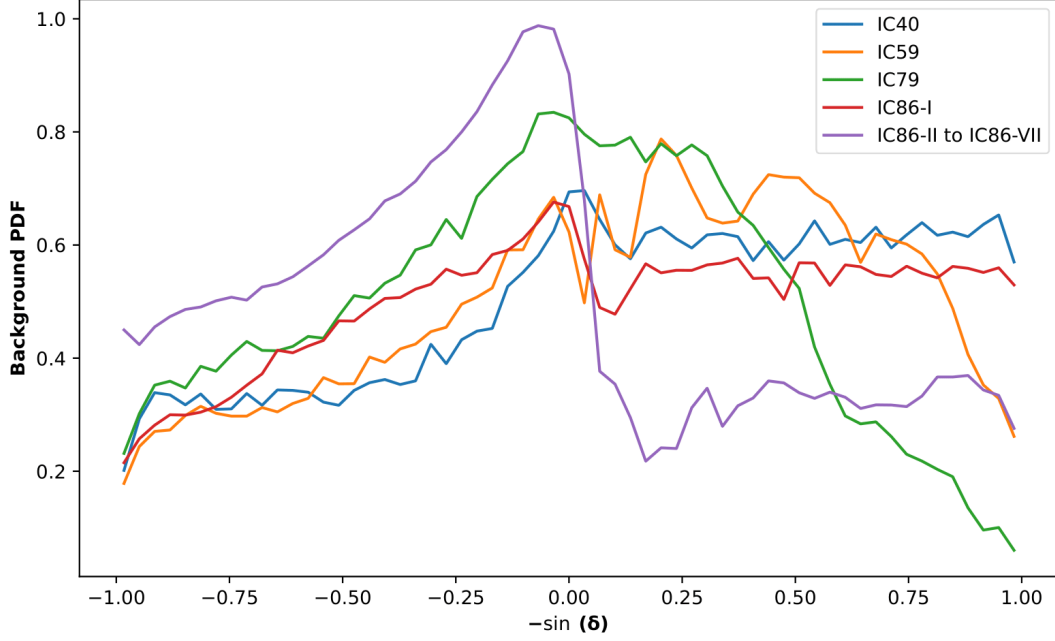


Figure 5.2: Background probability density functions (PDFs) for different IceCube seasons, as labeled. Shown are IC40 (1-year exposure, 36,900 events), IC59 (1-year exposure, 107,011 events), IC79 (1-year exposure, 93,133 events), IC86-I (1-year exposure, 136,244 events), and IC86-II to IC86-VII (6-year exposure, 761,143 events).

as follows:

$$\omega_{acc,j} = \sum_i^k T_k \times \int A_{eff}(E_\nu, \delta_j) E_\nu^\Gamma dE_\nu, \quad (5.11)$$

All terms in the above expression have the same meaning as in Eq. 5.7. For our analysis, we choose the spectral index $\Gamma = -2.53$.

The expected number of signal events coming from the source (n_s) is then given by:

$$n_s = \frac{\sum_j \omega_{model,j} \hat{n}_{sj}}{\sum_j \omega_{model,j}} \quad (5.12)$$

where

$$\hat{n}_{sj} = \sum_i^k T_k \times \int A_{eff}(E_\nu, \delta_j) \frac{dF}{dE_\nu} dE_\nu, \quad (5.13)$$

where $\hat{n}_{sj}$ is the number of signal events coming from magnetar j and $\frac{dF}{dE_\nu}$ is the expected neutrino spectrum from the source, which is defined in Eq. 5.6. Unlike single-source analysis, where n_s is obtained by maximizing TS for a single source,

here n_s can be deterministically evaluated from $\frac{dF}{dE_\nu}$.

Similarly to single-source analysis, we now plug the expressions for B_i and S_i from Eq. 5.9 and Eq. 5.10, respectively in Eq. 6.4, to determine TS . Similarly to [PD24], we show the plot for TS as a function of $E^2 \frac{dF}{dE_\nu}$, evaluated at 100 TeV. This plot of TS can be found in Fig. 5.3. We can see that the TS value is close to 0 and then gradually declines for $E^2 \frac{dF}{dE_\nu} \gtrsim 6 \times 10^{-6} \text{GeV cm}^{-2} \text{sec}^{-1}$. Therefore, we do not see a statistically significant excess using a stacked analysis. We calculated the 95% c.l. stacked upper limit on the differential neutrino flux by finding the X-intercept corresponding to $TS=-3.84$, which is equal to $8.8 \times 10^{-16} \text{GeV}^{-1} \text{cm}^{-2} \text{sec}^{-1}$.

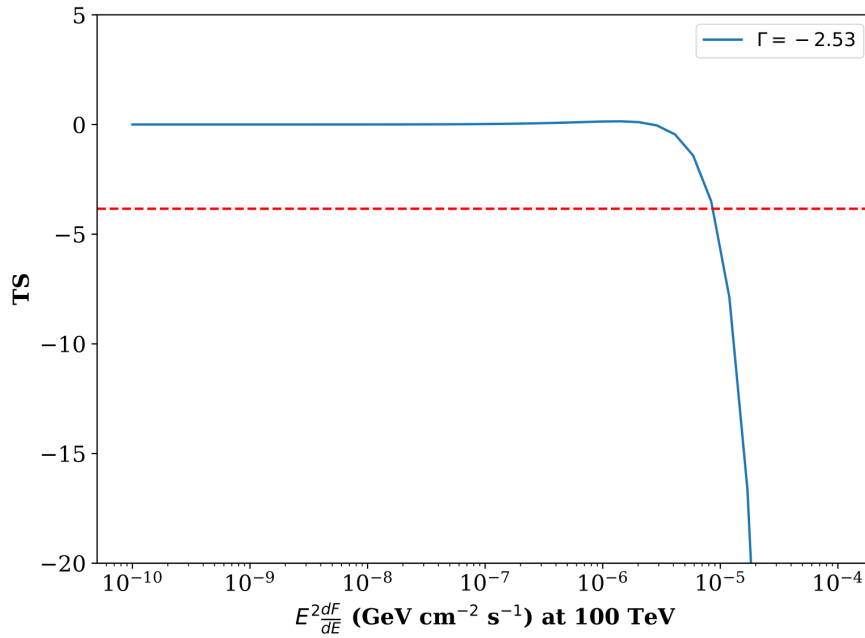


Figure 5.3: Plot of test statistic (TS) as a function of the total neutrino flux for 37 magnetars using IceCube 10-year muon-track data. In this analysis, we assumed $\omega_{\text{model}} = 1$ and $\Gamma = -2.53$. The dashed horizontal line corresponds to $TS = -3.84$, from which the 95% confidence level upper limit can be obtained from the X-intercept. The stacked 95% confidence level upper limit on the differential neutrino flux is $8.8 \times 10^{-16} \text{GeV}^{-1} \text{cm}^{-2} \text{s}^{-1}$.

Magnetars	RA (°)	Decl. (°)	n_s	$TS_{\max}$	Upper limit	ν Flux limit ($\text{erg}^{-1} \text{cm}^{-2} \text{sec}^{-1}$)
CXOU J010043.1-721134	15.17	72.17	8.94	0.94	34.8	5.3×10^{-12}
4U 0142+61	26.59	61.75	0.33	0.0008	26	3.7×10^{-12}
SGR 0418+5729	64.64	57.53	0.0	0.0	21	2.9×10^{-12}
SGR 0501+4516	75.27	45.27	0.0	0.0	16.5	2.2×10^{-12}
SGR 0526-66	81.50	66.05	2.44	0.09	23.5	3.4×10^{-12}
1E 1048.1-5937	162.52	59.87	0.0	0.0	12.1	1.7×10^{-12}
1E 1547.0-5408	237.72	54.29	1.098	0.01	25.4	3.4×10^{-12}
PSR J1622-4950	245.68	49.81	2.81	0.063	28.3	3.8×10^{-12}
SGR 1627-41	248.96	47.57	0	0	14.6	1.9×10^{-12}
CXOU J164710.2-455216	251.79	45.86	3.06	0.07	28.6	3.8×10^{-12}
1RXS J170849.0-400910	257.19	40.11	4.14	0.13	30	4.0×10^{-12}
CXOU J171405.7-381031	258.52	38.15	0.0	0.0	21.2	2.8×10^{-12}
SGR J1745-2900	266.41	28.99	2.29	0.08	23.8	3.1×10^{-12}
SGR 1806-20	272.16	20.38	0.0	0.0	15.2	1.9×10^{-12}
XTE J1810-197	272.46	19.70	0.0	0.0	17.2	2.2×10^{-12}
Swift J1818.0-1607	274.51	16.10	0.0	0.0	21.1	2.7×10^{-12}
Swift J1822.3-1606	275.57	16.05	24.19	3.92	0.2	2.5×10^{-14}
SGR 1833-0832	278.43	8.51	0.0	0.0	20.9	2.7×10^{-12}
Swift J1834.9-0846	278.71	8.73	0.0	0.0	13.2	1.7×10^{-12}
1E 1841-045	280.33	4.93	6.06	0.34	32.2	4.2×10^{-12}
3XMM J185246.6+003317	283.19	0.55	30.20	5.34	4.1	5.8×10^{-13}
SGR 1900+14	286.80	9.32	14.94	1.51	44.3	5.8×10^{-12}
SGR 1935+2154	293.73	21.89	16.35	2.18	43.5	5.6×10^{-12}
1E 2259+586	345.28	58.87	0.0	0.0	24.8	3.5×10^{-12}
SGR 0755-2933	118.92	29.53	16.38	2.44	43	5.6×10^{-12}
SGR 1801-23	270.24	22.92	7.46	0.66	31.3	4.0×10^{-12}
SGR 1806-20	272.04	20.61	0.0	0.0	12.7	1.6×10^{-12}
AX J1818.8-1559	274.71	15.97	0.0	0.0	19	2.4×10^{-12}
AX J1845.0-0258	281.22	2.91	0.0	0.0	26.6	3.6×10^{-12}
SGR 2013+34	303.48	34.33	0.0	0.0	23.1	3.0×10^{-12}
PSR J1846-0258	281.60	2.95	2.20	0.02	34	4.6×10^{-12}
GRB 231115A	130.75	73.5	0	0	14.4	2.2×10^{-12}
GRB 200415A	11.87	-25.02	0	0	9.8	1.4×10^{-10}
GRB 051103	148.14	68.84	0	0	16.1	2.4×10^{-12}
GRB 070201	11.07	42.3	0	0	16.2	2.1×10^{-12}
GRB 070222	205.53	-26.87	0	0	8.5	1.3×10^{-10}
GRB 180128A	12.3	-26.1	4.93	0.62	21.5	3.0×10^{-10}

Table 5.1: Results from spatial coincidence analysis between IceCube neutrinos and magnetars. The last three columns denote the number of observed signal events (n_s), the maximum observed significance ($TS_{\max}$) as described in Eq. 6.4, and 95% c.l. upper limit on the number of signal muon-induced neutrino events from magnetars (column 6), respectively. The last column shows the differential neutrino flux limit evaluated at neutrino energy of 100 TeV for a spectral index of -2.53. The flux limit can be evaluated for any other neutrino energy according to Eq. 5.6.

Chapter 6

Spatial coincidence between IceCube all sky point neutrino flux and Fast Radio Burst

Fast Radio Bursts (FRBs) are brief, millisecond-duration pulses of radio waves believed to originate from cosmological distances. These transient events remain one of the intriguing mysteries in modern astrophysics. The first known FRB was detected on July 24, 2001, using the 64-meter Parkes radio telescope in Australia. Designated FRB 20010724, the burst occurred roughly 3 degrees from the Small Magellanic Cloud and exhibited a peak flux density of ≥ 30 Jy at 1.4 GHz, with a temporal width of about 5 ms. The dispersion measure (DM) of 275 pc cm^{-3} significantly exceeded the predicted Galactic contribution of 25 pc cm^{-3} along that line of sight, based on the NE2001 model of the Milky Way's electron density, suggesting an extragalactic origin.

Until 2016, all detected FRBs appeared as single, non-repeating events. This changed with the discovery of FRB 20121102, the first known repeating source, identified by the 305-meter Arecibo radio telescope. It produced multiple bursts with a consistent dispersion measure, confirming its repetitive nature. Observational data now support the classification of FRBs into two categories: repeaters and apparent non-repeaters.

6.1 Data set

The publicly available IceCube all-sky points-wise catalogue contains 1,134,450 neutrinos spanning a 10-year time span from 2008–2018. Each neutrino event is provided

with the information of right ascension (RA), declination (delta), reconstructed muon energy, error, and detector zenith and azimuth angle. The CHIME FRB catalogue contains 491 FRBs out of which 18 are repeaters. These bursts have been detected between July 2018 and July 2019, and the frequency range is from 400–800 MHz. Since their observational period is different, there is no temporal overlap between the IceCube and CHIME FRB catalogue. CHIME FRB provided 54 attributes of each FRB. This catalogue covers mostly from the northern hemisphere and only 8 FRBs which is less than 0 declination.

6.2 Analysis and Results

For our analysis, we are using the unbinned maximum likelihood ratio method (See [PD22] and references therein, and first proposed in [BDD⁺08b]). We use neutrinos within a declination of $\pm 5^\circ$ of the FRBs. For a dataset containing N events, if n_s is the number of signals attributed to the FRBs, then the likelihood of the entire dataset is given by:

$$\mathcal{L}(n_s) = \prod_{i=1}^N \left[\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \right] \quad (6.1)$$

where S_i is the signal PDF and B_i is the background PDF. Here the signal PDF is given by:

$$S_i = \frac{1}{2\pi\sigma_i^2} e^{-(|\theta_i - \theta_s|)^2 / 2\sigma_i^2} \quad (6.2)$$

where $|\theta_i - \theta_s|$ is the angular distance between the neutrino and the FRB, σ_i is the angular uncertainty in the neutrino position, in radians. The background PDF can be obtained from the solid angle within the declination (δ) of $\pm 5^\circ$ around each FRB and is given as follows [BDD⁺08b]:

$$B_i = \frac{1}{\Omega_{\delta \pm 5^\circ}} \quad (6.3)$$

We then evaluate $\mathcal{L}(n_s)$ for different values of n_s , and designate the value of n_s , for which this likelihood is maximized as $\hat{n}_s$. Now to ascertain the significance of the signal, we define the test statistics (TS) based on the likelihood ratio, with respect to the null hypothesis in the absence of a signal, ($\mathcal{L}(0)$) as follows:

$$TS = 2 \log \frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(0)}. \quad (6.4)$$

For the null hypothesis, TS behaves like a χ^2 distribution for one degree of freedom [Wil38]. The detection significance statistic can be obtained from $\sqrt{TS}$. This statistics is also used in gamma-ray astronomy (eg. [MD24]). For a statistically significant detection of $> 5\sigma$, $TS > 25$. For FRBs, we plot the distribution of TS for the null hypothesis, which behaves as the distribution of χ^2 ???. Therefore, the observed signal events are consistent with the background, and there is no evidence of any spatial association between FRBs and IceCube neutrinos, which are observed as muon tracks. Consequently, we calculate the 95% c.l. upper flux limits by calculating the value of n_s for which $\Delta(TS) = -3.84$ [Z+20], where $\Delta TS = TS - TS_{max}$.

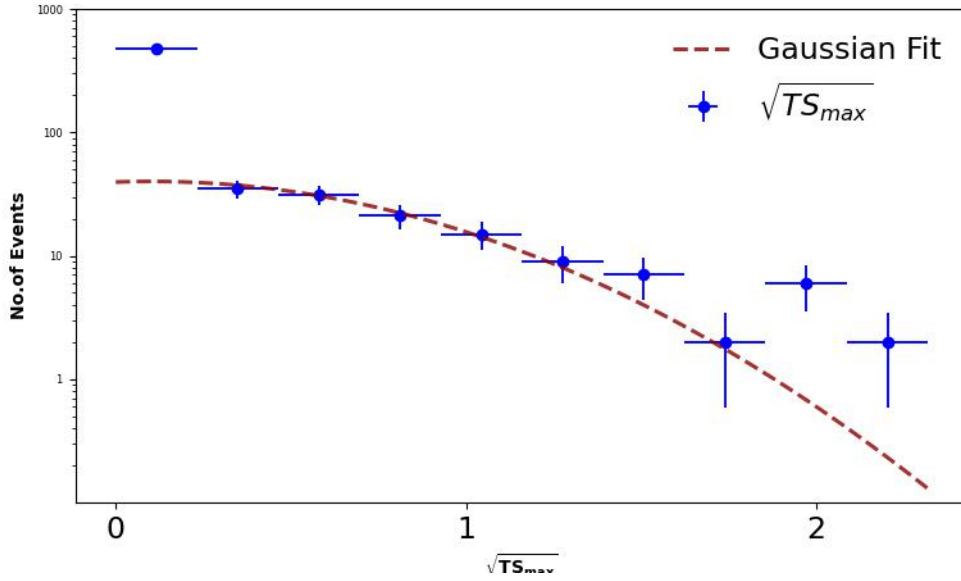


Figure 6.1: Distribution of TS for null hypothesis using FRB

Chapter 7

Conclusion and Future Directions

This thesis aims to search for the spatial coincidence between different Galactic sources such as gamma-ray bright red dwarfs, magnetars, and fast radio bursts, with high-energy neutrinos detected by the IceCube Neutrino Detector from 2008 to 2018. We analyzed our data using the unbinned maximum likelihood method. In each case of our search, we did not find any significance greater than 5σ . And a stacked analysis conducted for magnetars also showed a similar result. Hence, we conclude that none of the Galactic sources we observed contributed to the IceCube neutrino flux.

As a future direction, the upcoming IceCube upgrade and the IceCube-Gen2 will improve sensitivity and angular resolution. With a large volume and improved detection capabilities, IceCube-Gen2 will increase the likelihood of identifying weaker or more distant neutrino sources.

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